



NMRduino: A modular, open-source, low-field magnetic resonance platform

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ABSTRACT

The NMRduino is a compact, cost-effective, sub-MHz NMR spectrometer that utilizes readily available open-source hardware and software components. One of its aims is to simplify the processes of instrument setup and data acquisition control to make experimental NMR spectroscopy accessible to a broader audience. In this introductory paper, the key features and potential applications of NMRduino are described to highlight its versatility both for research and education.

1. Introduction

Over the past 80 years, Nuclear Magnetic Resonance (NMR) has evolved to become one of the most powerful analytical techniques available to chemists and materials scientists, able to provide detailed information about the molecular composition, structure, dynamics, and interactions in materials such as polymers, liquids, amorphous solids and crystals. Over these decades, particularly helped by the commercial era of NMR spectrometers that followed developments in pulsed-radiofrequency (rf) spectroscopy in the 1960s–2000s [1–4], NMR instrumentation has also become highly sophisticated in the unending effort to achieve higher sensitivity and resolution [5,6]. For example, a “standard” NMR-equipped laboratory for metabolomics, structural biology or drug discovery research will nowadays comprise an ultrahigh-field (> 15 T) magnet [7] equipped with a liquid-nitrogen-cooled detection coil [8] and perhaps infrastructure for hyperpolarization as well [9,10].

The historical evolution of NMR instruments should not, however, be viewed exclusively as a process of moving to progressively higher sensitivity and complexity. Nor should instrumentation from the past be overlooked as old technology of little practical relevance in the present day [11]. NMR is full of niches. Relatively low-tech and low-field implementations of NMR are of value in many applications outside the laboratory [12], including cultural [13,14], environmental [15–18] and industrial [19] studies where portability of instruments is of concern. In education, a relatively inexpensive, non-daunting spectrometer operating somewhere in between 1 T [20] and the Earth's field range [21–24] can facilitate the learning of basic NMR principles in parallel with textbook reading or classes. In other areas, such as

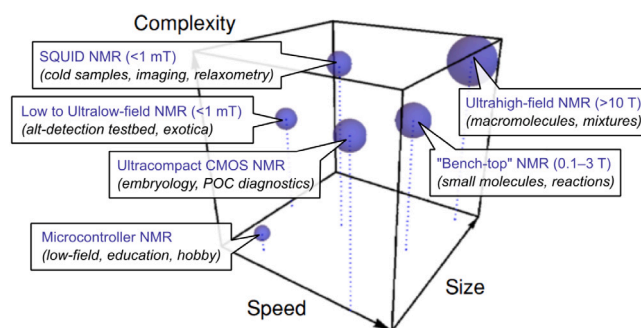


Fig. 1. Heuristic classification of NMR spectroscopy hardware according to different priority variables: speed (detectable Larmor frequency range), size and complexity of infrastructure, which is represented by the blue-colored bubbles. The bubble size loosely indicates the relative popularity of each technique.

embryology [25,26] and ultraportable point-of-care chemical diagnostics [27–29], miniature NMR detectors are in development on mm-scale monolithic CMOS substrates [30–32] containing all radiofrequency transmit-receive electronic circuitry and a microcoil resonator for signal pickup. One can summarize these niches as different optimizations of speed (which is directly proportional to the operating magnetic field, or Larmor frequency), size and complexity of the NMR experiment, roughly in that order of priority, as summarized graphically in Fig. 1.

In this paper, we discuss the unconventional scenario where an NMR instrument is designed unsparingly for minimum complexity.

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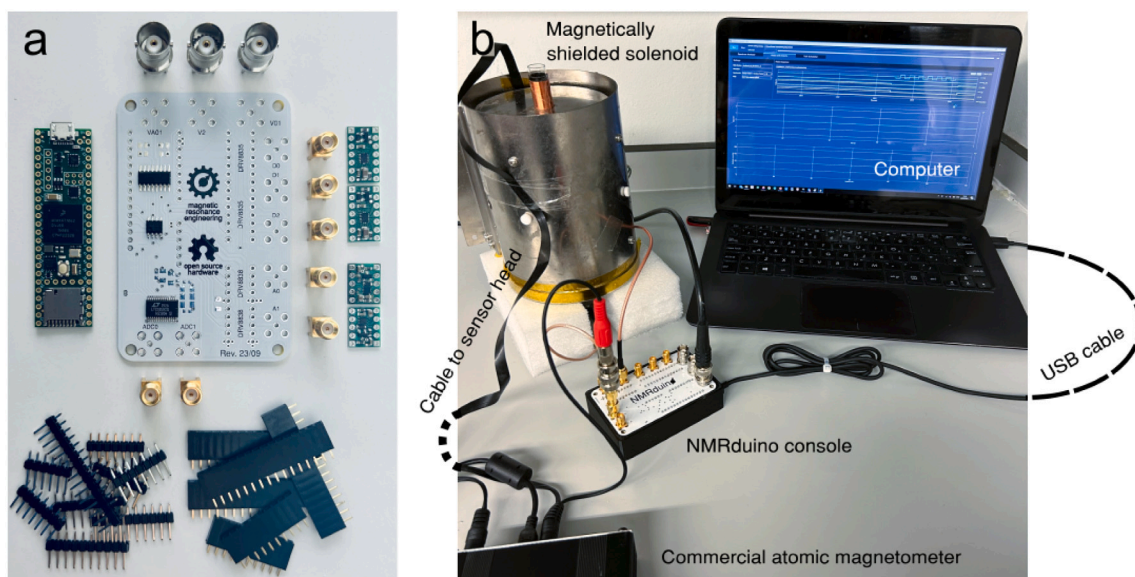


Fig. 2. Plug-and-play NMR using NMRduino: (a) the easy-to-assemble hardware components of NMRduino's current version. These comprise a 2-layer PCB measuring 83 mm by 54 mm — center, Teensy 4.1 MCU — left, H-bridge PCB modules — far right, coaxial cable connections plus 2.54-mm-pitch jumper connections — top and bottom; (b) a typical system-level implementation of NMRduino, which involves connecting the central PCB to a computer's USB port, a bench-top DC power supply (not shown), transmit coils, and pickup sensor of the NMR signal — in this case a commercial magnetometer sensor.

We define ‘complexity’ in this case as the monetary and time cost of making the instrument from mass-produced commercial parts, across all processes from electronic component sourcing and system assembly to programming and calibration, plus additional efforts made to ensure robustness and stability. Potential advantages of such a system may be a lowered entry barrier to NMR beginners or those users who wish to ‘hack’ (i.e., customize) hardware, plus a low cost, and high repair-ability or disposability. In practice, we have now realized such a system by leveraging the availability of widely supported mass-producible parts from three main areas: (1) low-cost modular electronic components, such as analog-to-digital converter modules for data acquisition and stepper motor drivers for current switching; (2) optical magnetometers, as a highly sensitive means of low-frequency signal detection; (3) microcontrollers compatible with widely supported open-source toolchains for overall system control. These components are interfaced with one another on a single printed circuit board, see for example Fig. 2, inexpensively produced and assembled by a one-stop-shop fabricator. The potential downside of a simplicity-prioritized approach to NMR is a constrained experimental capability, where in our setup, the accessible range of nuclear Larmor frequency is limited to below 100 kHz. However, although this frequency range is somewhat restrictive (for example, it does not enable the resolution of NMR chemical shifts for commonly studied nuclei, except perhaps for ^{129}Xe [33]), it still offers ample opportunities for alternative NMR experiments that yield other forms of resonance dispersion.

We have named our simplified NMR spectrometer platform ‘NMRduino’ [34] by association to (but we stress, not endorsement by, or affiliation with) the open-source ‘Arduino®’ microcontroller ecosystem [35] that its design draws upon extensively. The philosophy of Arduino revolves around making technology accessible to everyone, regardless of their background or expertise [36]. It achieves this by emphasizing open-license hardware and software tools with minimum up-front learning curve so that users can tinker, prototype, and focus on building their own electronic projects [37]. Some of the projects built by Arduino users in recent years do already include control and data acquisition systems for magnetic resonance, some handling radiofrequency (RF) pulse gating and DC field cycling in high fields (0.1–0.5 T) [38,39] while others operate near-Earth’s field, directly synthesizing the pulse waveform as well as digitizing the NMR signal [22,40–43]. Common to all of these is that the system-level control

code is simplified remarkably compared to traditional microcontroller programming [44]. So far, however, none of these projects have been widely adopted; in our view because of the abandonment of Arduino principles by their creators. In contrast, in the hope that NMRduino should become widespread as a magnetic resonance tool that is community supported, we aim for the platform to provide high accessibility, transparency, versatility and affordability.

The structure of the paper is as follows. We first present an overview of how NMRduino is constructed and may be operated by users. Basic introductory examples are then provided for users to follow who are either new to the NMR field or to operating an NMR spectrometer at a primitive level. A further set of examples follows, to illustrate the scope of the system in demonstrating advanced textbook principles of magnetic resonance that are widely practiced, such as spin echoes, heteronuclear pulse sequences [45] and hyperpolarization. Other use case scenarios for advanced users can be found in some of the authors’ previous publications, such as in field-cycling relaxometry [46,47], composite pulse benchmarking [48], spin decoupling in the nT to μT NMR regime [49,50], and many more topics [51–53] using preliminary versions of the NMRduino. Technical details of programming, control loops and limits on timing, data, and other input/output parameters are provided for reference purposes in the Appendix.

2. Conceptual definition of NMRduino

NMRduino encompasses both the hardware and software parts of a spectrometer system that is based on principles of modularity, transparency, straightforward plug-and-play functionality and community support. Implementations of NMRduino are designed to be user-ready, yet flexible enough to allow users to create their extensions or enhancements and contribute them back to the overall project [34,54]. To support development, NMRduino software is published as free to run, view, modify, and distribute under open-source licenses such as the GNU General Public License version 3. NMRduino hardware designs are released under a Creative Commons Attribution ShareAlike (CC-BY-SA) license; all parts of NMRduino therefore qualify as “open source”. Both mentioned licenses stipulate free commercial use. However, as a restrictive condition, these licenses also stipulate that any products dependent on or derived from the original work must release their entire source codes under the same or compatible licenses. Detailed license information is available on all download platforms [54].

3. Technical description of NMRduino

One can think of NMRduino as a hardware and a software ‘layer’ that is built on top of Arduino-compatible elements to enable and run the specific tasks of an NMR spectrometer. In common with Arduino, these high-level layers store all nonessential technical detail away from the user front end.

3.1. Hardware elements

The main hardware ‘layer’ of NMRduino is a carrier printed-circuit-board (PCB) on which modular components are mounted and connected to the computer via a universal serial bus (USB); the ‘backend’ of this layer is a pattern of PCB wiring traces that connects the components in a standardized way. In the current version, a credit-card sized (85 mm by 54 mm) PCB footprint is chosen to be visually appealing and compact while not being too crowded with components, as shown and described in Fig. 2a. A few other peripherals including electromagnetic coils, a low-voltage DC power supply and a probe are then connected to the PCB produce a complete spectrometer system; one such implementation is shown in Fig. 2b. A brief description of each component is given in the following sections. For advanced technical details we refer the reader to the Appendix, as well as an online repository containing the data sheets for the individual modular components [54].

The hardware is chosen so that, where possible, replacement part modules can be bought and plugged in to avoid the need to work/repair at the level of individual electronic components.

3.1.1. Microcontroller

The timing control and data-acquisition performance required to perform pulse-acquire NMR spectroscopy in a moderate field strength (> 1 T) can be met using the programming flexibility of *field-programmable gate-array* (FPGA) controllers [55–59]. While highly powerful and in principle very precise, FPGAs used in past open-source NMR systems were inherently complex to program and optimize, requiring a learning curve that would put off most potential tinkerers. A simpler alternative to control hardware where speed is not the ultimate requirement is a *microcontroller* or MCU [60]. Due to lower speed, microcontrollers are lower cost and more widely available, and owing to the efforts of open-source initiatives – principally Arduino – they have integrated development environments (IDEs), libraries, plus extensive community support resources that makes them easier for beginners to work with. For these reasons, the core of NMRduino is (presently) a microcontroller. Having said that, accessible IDEs and modular boards for FPGAs are now becoming available, e.g. the PYNQ platform with python support, and we hope accessible NMR systems will be based on these too in the future.

NMRduino uses a version of the ARM Cortex-M microcontroller known as the Teensy 4.1 [61] developed by PJRC (Paul Stoffregen’s Teensyduino). With a 32-bit processor architecture based on the IMXRT1062 chip (NXP Semiconductors N. V.) and a clock speed of 600 MHz, the Teensy 4.1 is an extremely high-performance MCU by current standards, and also provides generous general-purpose input/output (GPIO) pins and digital interfaces, e.g., inter-IC-sound – I2S, serial peripheral interface – SPI, and USB for connectivity. Another distinguishing feature of Teensy boards is their high compatibility with Arduino’s IDE and code libraries, allowing developers to leverage a large existing codebase as a resource. This and a compact size (18 mm by 62 mm PCB footprint) makes Teensy a popular choice of MCU among hobbyists and professionals alike.

At the time of writing, Teensy 4.1 boards may be purchased from a number of retailers including most major electronics component suppliers, online stores, and directly from PJRC. The Teensy is supplied with 0.1-in-pitch male header pins, which should be soldered to the PCB, and then plugged into the carrier PCB as shown in Fig. 2a. A USB micro-B to standard-A cable should then be used to connect the Teensy

to a computer, for power supply and programming of its firmware via a one-click upload within the Arduino IDE. The firmware is an Arduino-language-based C program that handles all tasks related to running the NMR experiment including

- waiting for, reading and interpreting USB-port commands (such as the pulse sequence instructions) sent from the computer at the start of an experiment;
- execution of the pulse sequence instructions in hardware;
- NMR-signal data acquisition and storage;
- exporting NMR signal data via the USB port for storage on the computer.

Firmware source files as well as instructions for upload to an off-the-shelf Teensy are provided on the repository [54]. Once the firmware has been uploaded, the MCU is ready to use; if desired, the user does not need to have any further involvement with the Arduino layer.

3.1.2. Current gating

Where an NMR experiment requires magnetic fields to be generated in electromagnetic coils, a significant electric current ranging from tens of mA to several amps must be sourced. For example, a one-layer solenoid made from 0.2-mm-diameter wire has a drive-current-to-field ratio of around 50 mA/mT, while a single-loop Helmholtz coil of radius 5 cm has a ratio of around 10 A/mT. A microcontroller cannot directly supply currents even in the tens of mA range, due to power limitations. However, it can serve as a precise controller for external current-supply circuitry through logic lines.

Teensy 4.1 offers the capability to simultaneously switch up to 32 logic lines within a single CPU clock cycle. In the current version of NMRduino, we utilize a subset of these lines (only 10) to control a total of five high-current bipolar outputs, consisting of three DC output channels and two AC channels. We refer to these as D0, D1, D2 and A0, A1 respectively.

The NMRduino produces bidirectional current control using H-bridge circuits. In brief terms, an H-bridge is a configuration of four MOSFET switches whose arrangement (at least on a circuit diagram) is a 2×2 matrix of vertical lines that resembles the letter ‘H’ when combined with a dividing horizontal line to represent the position of the electrical load, namely the coil (see Fig. 3a). The direction of the current across the load can be reversed by closing the switch contacts either (i) top-left and bottom-right, (ii) or bottom-left and top-right, or all switches can be open-circuit to leave the load electrically isolated, i.e., floating. Outside the world of NMR, H-bridge switches are extensively used in robotics and motorized vehicles to drive various types of DC motor coils, including stepper motors and brushless DC motors, where the dual polarity enables the motor to rotate both clockwise and counterclockwise and variable-duty-cycle pulsed-width-modulation (PWM) enables speed control. As such, H bridges are widely available as compact integrated circuits (ICs) with current rating of several amperes and μ s switching times where precise speed control is required. In the NMRduino we use a dual-channel H-bridge IC (Texas Instruments DRV8835 on Pololu carrier board) which can supply 1.2 A continuously to the D0/D1 channels and 2.4 A to D2 with PWM switching up to 250 kHz. The H bridge also includes built-in flyback protection diodes to prevent voltage spikes or damage to the circuit when switching the coil on or off. Each channel uses 2 logic lines, see Fig. 3b, one for on/off control (Dx_EN or Ax_EN) and the other for the current direction (or ‘phase’, labeled Dx_PH or Ax_PH). To establish the necessary connections between the Teensy and the corresponding H bridge inputs, the carrier PCB incorporates appropriate routing.

The two analog channels A0 and A1 allow current control beyond the simple bipolar on/off gating of channels D0/D1/D2 by having a DC voltage input to the H-bridge module that is software-controlled, also illustrated in Fig. 3. The control uses a precise digital-to-analog converter (DAC) in conjunction with a non-inverting operational amplifier, however, we stress, this is not the only method. For the Teensy

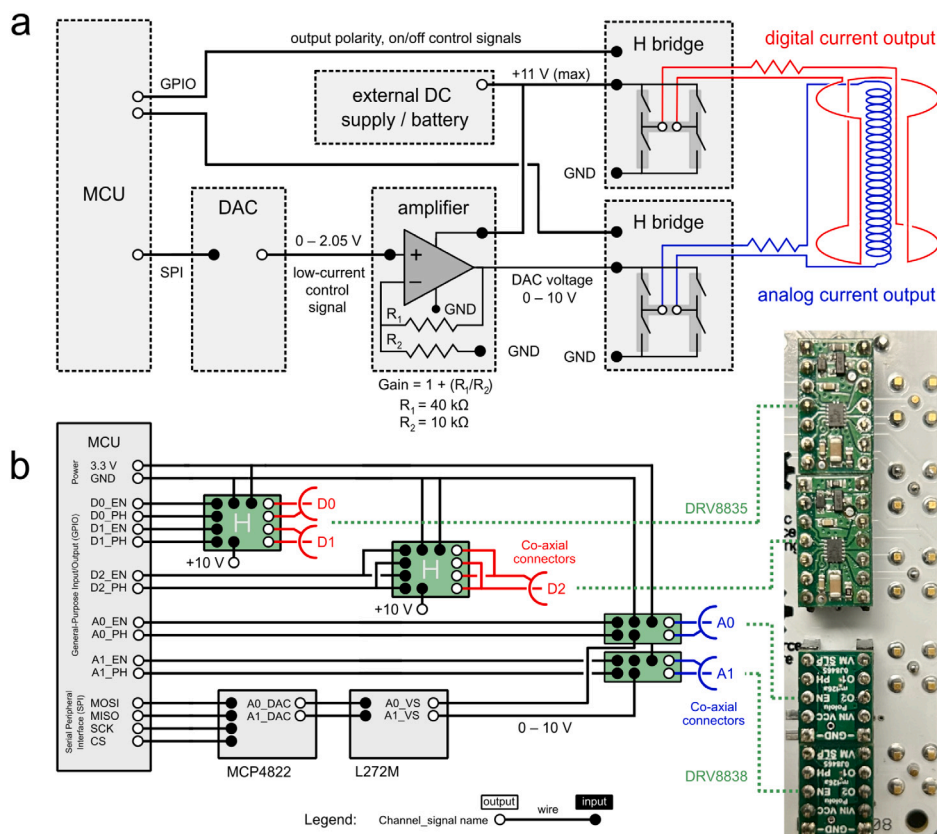


Fig. 3. The layout of NMRduino's switchable high-current outputs, (a) schematically and (b) showing the specific connections from the MCU. The external hardware comprises low-cost, preassembled H-bridge modules for bipolar on/off control diagram, plus digital-to-analog converters for amplitude control of the analog channels A0–A1.

4.1 microcontroller, we employ simple external components such as a dual-channel 12-bit DAC IC (Microchip MCP4822) connected to a single-ended dual amplifier IC (L272D) and two mono H-bridge circuit modules (DRV8838 on Pololu carrier board). An application of these independent current outputs is to drive two orthogonal field coils to produce a rotating or ‘circularly-polarized’ magnetic field. Again, the H-bridge modules are available commercially and simply require the soldering of 0.1-inch (2.54 mm) header pins and plugging into the carrier PCB. An alternative version of this circuit was presented in reference [48] using a Teensy 3.5 microcontroller; the Teensy 3.5 had a built-in DAC and the component MCP4822 was not required. However, the Teensy 3.5 is no longer produced or sold.

Depending on the needs of the experiment and power limits, the outputs D0/D1/D2 and A0/A1 can be used to drive accessory components of the NMR experiment such as laser controllers [51] or sample shuttling systems [46].

3.1.3. Signal input

The Teensy 4.1 features a built-in analog-to-digital converter (ADC) to interface with and record analog signal voltages. Rates of analog sampling can exceed MHz, although with the faster speeds the usable bit resolution is severely reduced; between 10 and 12 bits are typical. We find that higher bit depths are required for low-field NMR signal detection, and for this reason NMRduino uses an external ADC component such as the Analog Devices LTC1859 chip (8 channels, 16 bits, 100 kps maximum sampling rate). The LTC1859 has numerous attractive features for detecting spectroscopic signals, such as a bipolar input range (± 5 V or ± 10 V, giving resolutions of $76 \mu\text{V}$ or $152 \mu\text{V}$ respectively) and unipolar 5 V supply, the latter meaning the ADC can be powered directly from the MCU without additional voltage regulator components. Other features include software multiplexing of up to 8 input channels and built-in ± 25 V input protection, making it

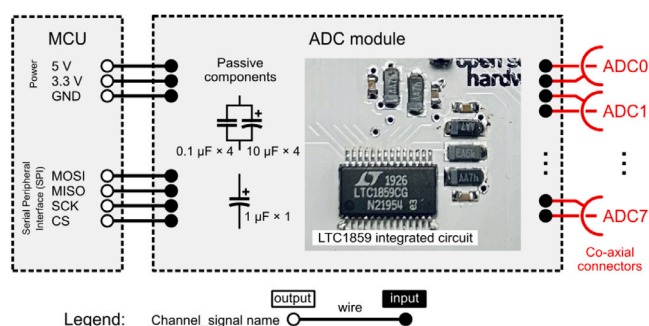


Fig. 4. The layout of NMRduino's ADC input module.

a forgiving ADC for beginners to work with. An alternative ADC with similar specification is the Analog Devices MAX1300/MAX1301, with 4/8 input channels respectively, 16-bit depth, a 5 V single-sided supply, ± 16.5 V input protection and throughput rates up to 115 kps.

Fig. 4 shows an example of how the LTC1859 ADC is interfaced with the MCU. Aside from power connections, a set of digital logic lines enables communication and control through the SPI or quad-SPI (QPI) protocol. The Teensy supports SPI data rates above 8 Mbit/s, which is fast enough to use the maximum sampling speed of the ADCs. The ADC is surface-mount-soldered directly on the NMRduino carrier PCB and is not intended to be user-serviceable. Here we note that although both LTC1859 and MAX1300/1 ADCs also share the same surface-mount-package (TSSOP-28) and therefore PCB solder-pad footprint, they are not exactly pin-for-pin compatible and the carrier PCB specific to each one should be used to ensure the correct serial, logic and power interfaces. As with the digital and analog output channels, the

PCB-specific port-to-pin mapping of the ADC input is coded within the Arduino firmware.

Each of the input channels of the ADC are broken out to a sub-miniature type A (SMA) socket. Users must connect these sockets to separate devices or circuits that detect a component of the nuclear magnetization according to their needs, e.g., Larmor frequency range, detection bandwidth and noise. Supported devices currently include vector atomic magnetometers or tuned RF induction coils with a preamplifier output. A brief review of these devices is given below.

- **Atomic magnetometers:** these are sensors that draw upon the magneto-optical properties of atomic ensembles such as alkali-metal vapors [62] or triplet noble gases [63] where the orientation and magnitude of an applied magnetic field can produce changes in atomic spin states populations that are measurable using light. They offer a novel avenue for optical detection of NMR in the low-frequency low-field realm, reaching sensitivities down to a few femtotesla ($< 10 \text{ fT}/\sqrt{\text{Hz}}$) typically operating in a background magnetic field of a few μT or lower, such as that provided by a magnetic shield (Fig. 2b). This suffices to measure ^1H precession signals in the Earth's field with a relative noise floor below $0.01 \text{ scan}^{-1/2}$, for example 1 mL of pure water (H_2O) after remote prepolarization at 1–2 T field gives a signal-to-noise ratio of several hundred in a single scan.

Of the several flavors of atomic magnetometer that currently exist [64,65], the most suitable one for a given NMR measurement scenario depends on the sensor size, geometry and frequency range of interest. Those based on zero-field (Hanle) resonance effects in alkali-metal vapors [66] can be miniaturized to thumb-sized sensor packages that are highly convenient for arraying but only cover frequencies from near-DC to a few hundred Hz. Alternatively, Faraday-rotation magnetometers have a much larger frequency range of up to several kHz [47], although with a larger apparatus footprint. Bell-Bloom and other types of RF magnetometers can address higher frequencies. Some of these magnetometers are now produced commercially [67,68] and have dual use, for example, the QZFM family of optically pumped magnetometers (QuSpin Inc., USA) can switch between different modes for kHz-band detection [69] as well as near-DC detection [70]. The main focus of these commercial prototypes is presently the market of noncryogenic magnetometers for magnetoencephalography [71] and microfluidics [72]. Their utility and simplicity means undoubtedly a wider range of cost-effective off-the-shelf products for NMR will become available in the future. An interesting study avenue is the DC detection of magnetic fields from the NMR sample along the axis of the magnetic bias field, which is inaccessible to inductive coil detection [50,53].

- **Inductive coils:** a wire coil is placed around the NMR sample, serving as a classical transducer between the precessing total magnetic moment of the nuclear magnetization and an alternating voltage in the coil conductor. Signal-to-noise scales roughly in proportion to frequency $^{3/2}$, all other factors being equal, including initial nuclear spin polarization.

An inductive coil is usually optimized at a fixed Larmor frequency for maximum sensitivity plus magnetic homogeneity, noise reduction and portability. This makes it difficult to use for NMR detection across variable fields especially at low frequencies (1 kHz to 1 MHz) where inductive components (besides the coil) must be added in the circuit each time to reach a different tuning or matching condition. Combined with the low raw materials cost and wide options for customization, for example differential coil setups [43], Q-switches [73], ultrahigh-Q resonators [74] and low-noise preamplifiers to minimize external interference, this type of detection hardware is mostly custom-made rather than off-the-shelf. Existing commercial products have been prepared for sale mostly as educational demonstrations such as the Terranova

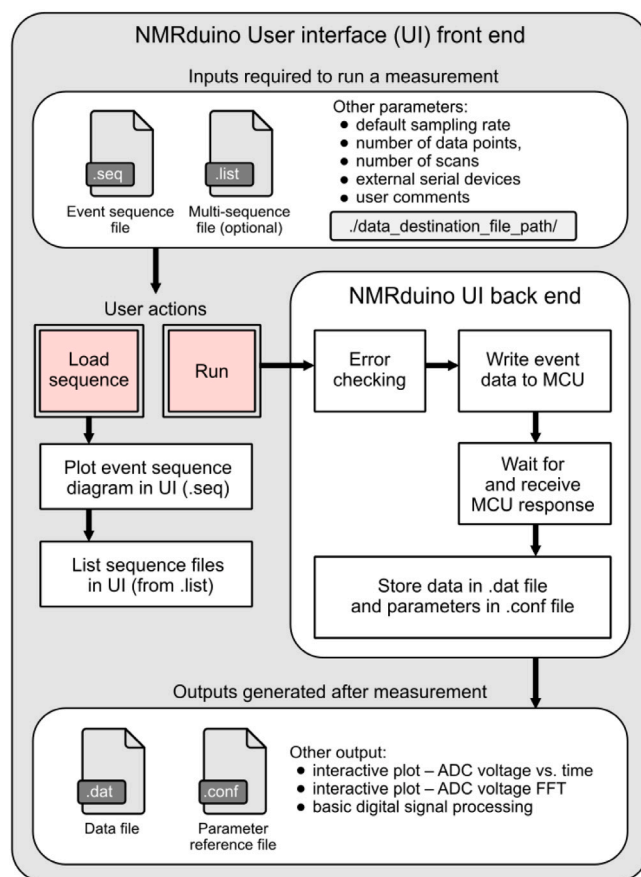


Fig. 5. Graphical representation of the NMRduino UI. A more detailed view can be found in Fig. 16 in Appendix.

Earth's field MRI system [75,76] (Magritek Ltd.), or as products for medical niches of NMR, such as hyperpolarized-xenon polarimeters that operate at mT fields [77–80].

3.2. Non-hardware elements

The software component of NMRduino is a multi-level structure designed to execute pulse-and-acquire event sequences with precise timing and afterwards store the measured data. As outlined below, the highest level is a graphical user interface (UI) that can be run on the desktop computer, laptop or single-board computer (e.g., Raspberry Pi) connected to the MCU USB. The Arduino firmware forms the lower level and determines the MCU control of the PCB peripherals. Additionally, NMRduino provides a simple and transparent format for arbitrary pulse programming.

3.2.1. User interface

The UI facilitates the task of instructing the MCU and other peripherals to execute the NMR experiment. Up front, users can specify a pulse program stored in the computer file system, a desired data storage directory, and then click a single button to start measurement: see Fig. 5. The UI back-end checks the pulse program for critical errors, and if error-free, then uploads the sequence to the MCU via USB and waits to receive acquired data, all while providing an update of the experiment's progress throughout. An expanded description of the back-end routine especially pulse program conversion (see Fig. 17) is given in the Appendix.

The UI is written using the C-based development framework Qt [81], which provides extensive libraries to implement a range of interactive

features. For example, data received from the MCU are graphically presented as a time-domain signal trace and a discrete Fourier transform spectrum. This enables users to visually inspect the data directly and intuitively interact with the plotted graphs to zoom in/out, scroll, switch between log/linear scales, and even select digital filters to enhance the data visualization, without the need for additional processing elsewhere. For users who wish to perform processing in external programs, each data file is generated and stored separately as a plain binary format (‘.dat’ extension) for efficient storage and easy retrieval. An accompanying configuration file (‘.conf’) is generated in the same directory to store all of the UI-editable experimental parameters. These parameters include data length, sampling rate, pulse event sequence filename, and the number of scans, among others. This ensures that all experiment-specific settings are preserved for future reference and data analysis. A local .conf file is also continuously updated within the UI’s home directory. In the event of a power cut or other termination of the UI program, this allows state of the system to be quickly restored, plus any ongoing measurements resumed.

3.2.2. MCU firmware

The primary function of the firmware is to interpret and map the pulse program into a sequence of hardware instructions. It operates as a finite-state machine with the following core states including: listening for USB commands; reading pulse program data from USB; parsing and executing programmed pulse/acquisition instructions event by event (line by line); writing acquired data to USB. The firmware utilizes high-level Arduino code macros for programming convenience, along with low-level, MCU-specific register read/write instructions and interrupts to achieve clock-precise timing. Additional information on the interrupt timer and live event parsing is provided in the Appendix, as illustrated in Fig. 18.

3.2.3. Pulse programming

From the electronics perspective, the NMR experiment is an orchestrated sequence of voltage switching and sampling events where the controller outputs a chronological list of instructions to its I/O. When dealing with a DC pulse for example, there are two main events: switching the H-bridge logic to (1) turn on the current output, then (2) turn the current off. Each of these events occurs at a specific time, referred to as the “time stamp”, for which the program should instruct the updated amplitude of voltage applied across the load. An AC pulse may have many more events, perhaps 10 or 100, to update the voltage amplitude at multiple points throughout the wave’s period.

NMRduino provides a framework to write down these voltage switching and sampling events in digital form on a computer, send them to the microcontroller, and ultimately break them down to actual I/O switching operations for execution with correct timing. This framework is not only flexible but has several advanced capabilities. For example, any multiple of the digital or analog channels can be simultaneously switched during one event, with one time stamp. It also handles sequences in which current switching events and data-point acquisition are interleaved with one another, e.g., windowed decoupling-acquisition, or Carr–Purcell–Meiboom–Gill (CPMG) echo train acquisition (see Section 4.4).

An example sequence for illustrative and test purposes is given in Fig. 6. It is stored in tab-separated ASCII format, though a specific “.seq” extension is used to denote within NMRduino that it is an event sequence. The first row of the file is a header that indicates the total number of events, and the names of the output channels assigned to the data columns below. Subsequent rows denote single events in chronological order from top to bottom, where the number in each column details the nature of the change to the input or output. We note that while the rows must be ordered chronologically, the columns can be given in any order albeit that the first column must contain the time stamp (in μs).

	19#	A1	A1_dac	D0	D1	ACQ	dt
2	1	0	0	1	0	0	0
3	3	0	0	0	0	0	0
4	4	0	0	-1	0	0	0
5	6	0	0	0	0	0	0
6	7	0	0	1	0	0	0
7	9	0	0	0	0	0	0
8	10	0	0	-1	0	0	0
9	12	0	0	0	0	0	0
10	20	0	0	0	0	2000	10
11	9999980	0	4095	0	0	0	0
12	9999990	-1	0	-1	0	0	0
13	9999991	0	0	-1	0	0	0
14	9999992	1	0	-1	0	0	0
15	9999993	0	0	-1	0	0	0
16	9999994	-1	0	-1	0	0	0
17	9999995	0	0	0	0	0	0
18	9999996	1	0	0	0	0	0
19	9999997	0	0	0	0	0	0
20	10000000	0	0	0	0	1000	20

Fig. 6. Example structure of a .seq file used by NMRduino to specify pulse-acquire event sequences. The left-hand column denotes the event time stamp in μs , while all other columns provide the event data. All columns are tab-separated.

The test sequence illustrates a set of events where the D0 output channel is switched on for 4 alternating square pulses of duration $2\mu\text{s}$ and separation $1\mu\text{s}$, starting at $1\mu\text{s}$ and finishing at $12\mu\text{s}$. The data value in the rows is decoded as follows: “1” represents a high output, while a “0” represents off and “-1” denotes high output in the opposite direction. Channels D0 and A1 are then switched on just short of 10 s later for a second quartet of pulses, this time of duration $1\mu\text{s}$ and separation $1\mu\text{s}$, starting at 9.999990 s and finishing at 9.999997 s . The A1 DAC value is set $10\mu\text{s}$ in advance of the first pulse to ensure settling of the DAC and stabilization of the H-bridge charge pump. Interspersed with these pulse events are data point collection periods, starting at $20\mu\text{s}$ (2000 points, 10 kps) and 10 s (1000 points, 5 kps). The sequence serves to check timing output on an oscilloscope and does not represent any particular NMR experiment.

NMRduino adopts the above structure of event programming for several reasons. The main one is memory efficiency: many common NMR sequences comprise a short (μs to ms) sequence of pulses with relatively large waiting delays (s) in between. Output-change events take only a small fraction of the file size to store compared to the entire waveform of each channel during the sequence, and this approach of dropping of redundant points in the pulse waveform means there is zero penalty for large delays between events. For instance, there is no penalty to increase the delay between pulses from 10 s to 1000 s in the example given in Fig. 6, because the number of events in the sequence would stay the same. Secondly, only I/O change data are transferred to the MCU, in order to minimize time over the relatively slow USB port. Finally, the event-change structure is more human readable and is more amenable to making quick edits or modifications to the sequence with a generic text editor.

Users can inspect the pulse sequence output using an oscilloscope, a suitable resistive load (e.g., a coil of wire with resistance $\gtrsim 4\Omega$, or a 50Ω terminator) and a few coaxial cables. The components are connected to the oscilloscope as shown in Fig. 7 to measure the differential voltage reading across the load, which because of the H-bridge circuit is electrically floating with respect to the NMRduino. A video clip of this procedure is given in the Supporting Information.

The oscilloscope display trace allows users to check that the sequence-specific voltage waveform is shaped correctly or otherwise needs to be modified, such as (i) by correcting a typographical mistake in the .seq file or (ii) by adjusting DAC output values to compensate for output non-linearity in the DAC or current amplifier. Additionally, users may determine bandwidth-related imperfections in the waveform, such

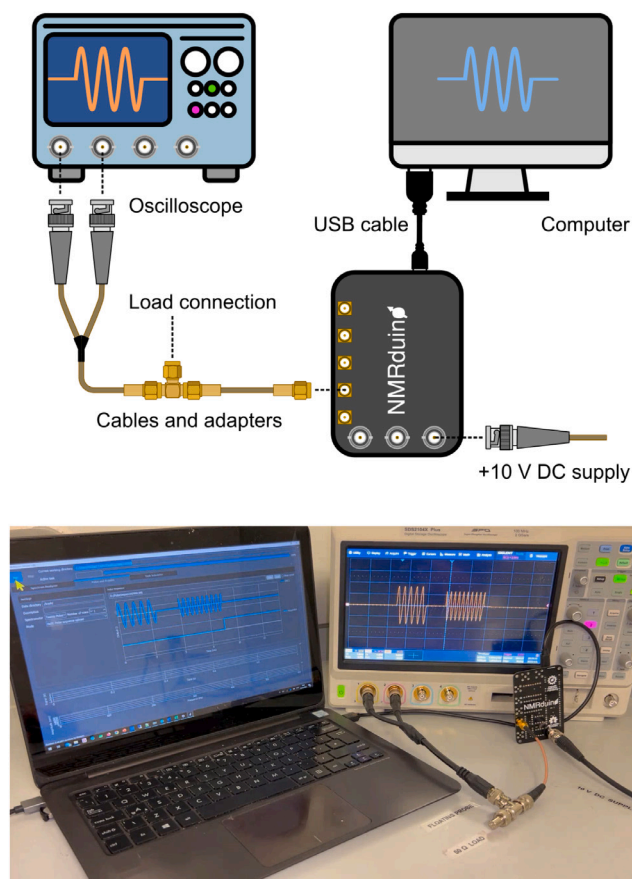


Fig. 7. Typical setup for testing NMRduino pulse-sequence output on a differential oscilloscope. The Y-junction in the co-axial cable connects the ‘+’ and ‘-’ wires of one floating bipolar-output to the two center pins of the oscilloscope input. The display shows the subtraction of the two input channel voltages.

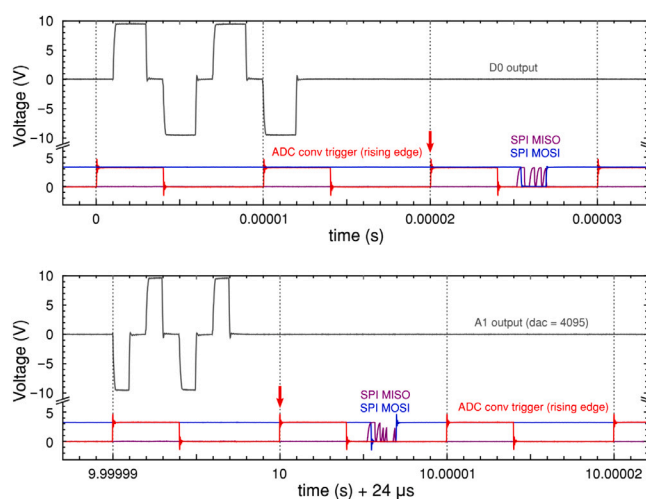


Fig. 8. Representative timing output of NMRduino for the multi-pulse test sequence given in Fig. 6. Measured using a 100 Mps oscilloscope, triggering on the rising edge of the first ADC conversion (red curve) to define the zero of the horizontal axis. The outputs A1 and D0 were each connected to a load coil of resistance 10 Ω .

as the pulse rise time and the fall time, the timing variability of sharp voltage transitions (jitter) and voltage overshoots and transients when switching current through inductive loads. An example measurement is given in Fig. 8. In these tests the outputs of the NMRduino are fed

into to a pocket-sized 30 MHz oscilloscope (Analog Devices: Analog Discovery 2), and both devices are connected to the same PC via USB. Locally speaking the sequence performs well with μ s timing precision, for instance the two sets of pulses near 0 and 10 s show pulse (90%) rise and (10%) fall times below 0.1 μ s. The 3.3 V logic outputs are included below to indicate the times of ADC triggering at every multiple of 10 μ s (red curve, rising edge) and data collection (blue/black curves, 8 MHz SPI master-out-slave-in/master-in-slave-out lines). Globally, there is a timing discrepancy of around -24μ s (2.4 ppm) near the next set of pulses at 10 s and this varies slightly by 3 to 5 μ s (0.3–0.5 ppm) when testing on different days, due to drifts in the laboratory temperature (~ 0.3 – 0.4 ppm/K around room temperature). The shift is attributed to the lower accuracy of the Teensy’s crystal oscillator (2400T, frequency accuracy 10–20 ppm) although measured result appears to be well within the specification because the expected drift at 10 s is closer to around 200 μ s. We ignore any systematic timing error introduced by the oscilloscope because its frequency accuracy is much better, specified as less than 1 ppm. A global timing inaccuracy of this kind is generally unimportant for NMR experiments in the low-frequency range. However, if better accuracy is desired, it should be possible to desolder the 24 MHz oscillator from the Teensy MCU circuit board and replace it either by (i) one that has a lower error, or (ii) an external frequency standard. Sub-ppm-error surface-mount oscillators can be bought individually and cheaply from most major electronics suppliers.

Some important limits of the event-sequence structure should be noted. The maximum length of the event sequence is limited by the on-board memory of the MCU to around 20,000 events, though improvements of around a factor of 2 may be possible through RAM sharing techniques or external RAM hardware if the user is willing to modify the MCU’s programming. The maximum event time stamp is also limited, due to the size of the timer counter (see subsection below), a 32-bit unsigned integer. This corresponds to $2^{32} \mu$ s or just under 1.2 h as the maximum length of the event sequence. For most NMR experiments this is unlikely to be a limitation, considering that nuclear spin relaxation times are typically on the order of s and at best on the order of a few minutes — and only in very exceptional cases hours. Finally, pulse and acquire events cannot presently share the same time stamp, though close timing is permitted. Acquisition must wait 1 μ s after a pulse, while pulses can only follow 1 extra sampling period after acquisition ($>10 \mu$ s). Again, these limits should not pose a problem for the majority of low-field, low-frequency NMR experiments.

Finally, for going beyond single-sequence measurements the UI also provides a convenient single-click functionality where as an alternative to selecting a single ‘.seq’ file path, users specify a separate file containing a list of file paths and the number of scan repetitions that should be performed for each one. This parent file, an example of which is provided in Fig. 9, is distinguished from the individual ‘.seq’ files by having a different extension: ‘.list’. Upon clicking the ‘Run’ button, the UI loads these file paths into a queue allowing them to be executed sequentially, one after the other, until the entire list is completed. The feature proves valuable to follow variations in an NMR signal across one or more experimental parameters, or implement pulse phase cycles. For instance, users can schedule a list of event sequences that have varying pulse length to measure a Rabi curve (see Section 4.2), or alternatively a varying delay interval between events to measure an evolution frequency or decay time constant of the NMR system.

4. Basic introductory examples

Here we outline simple experiments that may be useful to those who are learning about the practical details of NMR, and describe how these may be performed with NMRduino. Expert practitioners of NMR may wish to skip ahead to the more-advanced examples in Section 5.

1	4#	PulseSequencePath	NumberOfScans
2		/folder_1_path/sequence_1.seq	2
3		/folder_2_path/sequence_2.seq	2
4		/folder_3_path/sequence_3.seq	2
5		/folder_4_path/sequence_4.seq	2

Fig. 9. Structure of the .list file for multi-sequence experiments. All columns are tab-separated.

4.1. Larmor precession

Precession – literally and figuratively – is the pivot around which the entirety of NMR revolves. This is the phenomenon where the magnetic moment of a nucleus, a vector parallel to its spin angular momentum, rotates through time around the axis of an external magnetic field ($\mathbf{B}$). The spins' angular velocity (ω) is given by $\omega = -\gamma \mathbf{B}$, where γ denotes the nucleus-specific gyromagnetic ratio, whose value by this time has been precisely determined and listed in countless reference tables [82]. Experimental measurements of free precession in an NMR sample may therefore appear rather trivial. However, they can be one of the most educative demonstrations of how an NMR spectrometer actually works, from measuring intrinsic sample properties (textite.g., J couplings, transverse relaxation times) to important aspects and limits of hardware (textite.g., field-to-current ratio of coils, magnetic gradients, tuning response and sensitivity vs. frequency).

As discussed earlier, a few crucial components are needed, and electrically speaking all of these are capable of being sourced from or interfaced with NMRduino. These components must: (1) generate the $\mathbf{B}$ field; (2) detect a component of the rotating magnetic field and record it in digital form for analysis; (3) coherently tilt the magnetic moment of the spins away from alignment with the $\mathbf{B}$ axis to create a rotating magnetization. For these we return to the minimal setup described in Fig. 2b. For (1), the magnetic field is supplied by a solenoid: thin (30-gauge) enameled Cu wire is coiled around a carbon-fiber tube (15 mm outer diameter), and this is connected directly either to one of the NMRduino current channels or a low-voltage laboratory power supply (e.g., Hameg HM7042-5 or Twinleaf CSB). For detection, (2), an atomic magnetometer is placed on the outside of the tube with its sensitive axis pointing in the radial (y) direction to pick up the precessing magnetic field perpendicular to the z axis [47]. A resonant excitation pulse, for (3) is supplied through a saddle coil along the x axis, which we connect to the A0 channel of the NMRduino for its supply voltage. A few additional components are also added as indicated in Fig. 10a, namely a fluidic system involving two 1 mL syringes to shuttle the liquid sample between the detection region of the magnetometer, and a low-homogeneity but strong (~ 1 T) permanent magnet [83] for initial prepolarization. Push and pull action of the syringe plungers is controlled by a linear actuator (Glideforce GF23-120504-1-65 from Pololu Corporation, 100 mm stroke length), which can be driven directly by pulse-width modulation from the H-bridge output of the D2 channel according to instructions given in the event sequence file. The sample is shuttled through a 0.8 mm inner-diameter PEEK tube of length 15 cm. The thin tube results in a dead volume of $\sim 75 \mu\text{L}$, and therefore most of the polarized fluid reaches the detector.

Example NMR spectra measured with this system are shown in Fig. 10b. The fluid was a 1 mL solution of 9 M lithium chloride (LiCl) in water (H_2O), chosen to obtain precession signals from the uncoupled hydrogen (^1H , spin 1/2, 99.99% abundance, $\gamma_{^1\text{H}}/(2\pi) = 42.577 \text{ Hz } \mu\text{T}^{-1}$) and free lithium cation (^7Li , spin 3/2, 92% abundance, $\gamma_{^7\text{Li}} \approx 0.3886\gamma_{^1\text{H}}$) species. After the shuttling, a resonant $\pi/2$ pulse was applied to each spin species to excite the observable – nuclear magnetization in the xy plane. Optimal pulse lengths were found using the procedure described in Section 4.2.

The bias field was set to $B_z = 100 \mu\text{T}$ by adjusting the solenoid current to shift the ^1H resonance line to 4258 Hz, at which point the

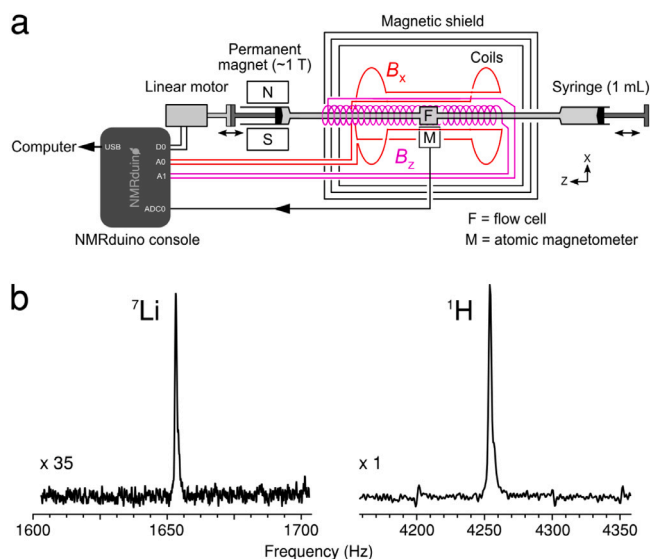


Fig. 10. Low-field (100 μT) NMR spectroscopy of uncoupled spin species using NMRduino: (a) Schematic experimental setup for remote prepolarization of a fluid in a low-homogeneity 1 T permanent magnet followed by shuttling and detection in a flow cell next to an atomic magnetometer. The flow cell is a cylinder of approximately 10 mm internal diameter and 13 mm height. The shielded part of the setup is roughly similar to that shown in Fig. 2b; (b) The Fourier transform of pulse-acquire precession signals measured for 1 mL, 9 M LiCl in H_2O . The spectrum on the left is an average over 20 scans while that on the right is from a single scan.

^7Li resonance was at around 1656 Hz or 0.3886 times the ^1H frequency as expected from the ratio $\gamma_{^7\text{Li}} : \gamma_{^1\text{H}}$. The final value of 13.33 mA current thus allows a field-to-current ratio of $\text{d}B_z/\text{d}I = 7.5 \text{ mT A}^{-1}$ to be inferred, and this compares well to the value expected from Ampere's law. These and other calibration settings may be stored by the UI and used in future experiments.

As a concluding point, we emphasize that each plot in Fig. 10b displays the result of separate measurements and not two regions of the same spectrum. This results from the relatively low bandwidth of the sensor (~ 250 Hz for a typical alkali-vapor magnetometer) and the need to tune the center frequency of the pickup to the Larmor frequency of interest, f_0 . Briefly, tuning is achieved by setting the background magnetic field of the magnetometer to $f_0/\gamma_{\text{alkali}}$ where γ_{alkali} is the gyromagnetic ratio of the alkali-metal atom, so that the nuclear spins in the sample and the alkali-metal atom spins in the detector precess at the same frequency. This is done using an extra z -axis coil, not shown in Fig. 10. For the Rb-vapor magnetometer used in this work, the gyromagnetic ratio used is $\gamma_{\text{Rb}}/(2\pi) \approx 7 \text{ kHz } \mu\text{T}^{-1}$ and therefore the bias field corresponds to about $0.6 \mu\text{T}$ for tuning to ^1H and $0.24 \mu\text{T}$ for tuning to ^7Li . The attraction of the magnetometer is that one can simply and quickly tune between these frequencies by switching the bias field. In comparison, the process of tuning a traditional inductive-pickup NMR coil is much slower and more involved; the circuit capacitance and inductance must be co-adjusted to tune to the resonance frequency and match the electrical impedance. Further detail on the magnetometer tuning procedure can be found in other works [47].

4.2. Rabi curves

In many spectroscopic techniques, NMR included, certain parameters of the system might not be observable 'directly' but may be discerned by techniques that measure their effect on other observables, for instance the amplitude of the precession signal. This is termed indirect detection, and an example carried out in almost all NMR experiments is the measurement of *Rabi oscillations*. These describe rotations of the nuclear magnetization vector away from the axis of the main solenoid field (z direction) by transverse field pulses (x direction).

The textbook reader may be familiar with explanations of the Rabi oscillation phenomenon that involve a resonant reference frame that rotates about Z at the applied RF frequency. In this article, we consider a simple and alternative model in the laboratory frame. Let the amplitudes of the transverse (x -axis) and longitudinal (z -axis) magnetic fields be denoted respectively as B_x and B_z , so that the total field is oriented at an angle $\theta = \arctan(B_x/B_z)$ to the z axis, in the xz plane. Assuming these fields are constant, the time to precess through an angle of π is $\tau_\pi = \pi/[\gamma(B_x^2 + B_z^2)^{1/2}]$ or $\tau_\pi \sim 1/(2f_0)$ for $|B_x| \ll |B_z|$.

If after time τ_π the B_x field is turned off, the precession signal will be proportional to $\cos(2\theta)$ because the spins have rotated by 2θ in the xz plane. Although small, this rotation can be compounded by applying the x -axis field in the opposite direction for a further time delay τ_π , so that after N square pulses of alternating sign and total duration $N\tau_\pi$ the overall accumulation is $2N\theta$, or $2NB_x/B_z$ in the limit of small θ , see for instance Fig. 11a. Thus, when θ is known, N can be determined to impart a particular large-angle reorientation on the spins, e.g., $\pi/2$ or π .

It is typical to determine θ by experimentally recording the amplitude of the precession signal as a function of pulse length (τ) and N . Invariably when starting out one finds that a first guess of τ is close to but not equal to τ_π , so the orientation of the magnetization vector after each rectangular pulse is not completely within the xz plane, but leaks out into the y direction. This situation corresponds to an off-resonance pulse, and leads to a lower amplitude of the observed signal through destructive interference (see Fig. 11a blue/black curves).

In Fig. 11b we illustrate results using a square wave current output from the A0 channel of the NMRduino to supply B_x . The coil was of a saddle geometry, 13 cm in diameter and 13 cm length wound with 6 layers of 0.3 mm diameter enameled copper wire. A shunt resistor was added in series to bring the resistance to 80 Ω ; the 10 V DAC output from the NMRduino therefore corresponding to a DC load current of 12.5 mA. The plots show the ^1H NMR signal amplitude dependence on the number of square wave pulses and the square pulse length τ for $^1\text{H}_2\text{O}$ in a field $B_z = 100 \mu\text{T}$. In terms of control implementation, separate .seq pulse sequence files were written for each value of τ and N , and these were queued in a single .list file, 1 scan per sequence.

The maximum signal occurs at 120 μs ($N = 7$, i.e., 3.5 square-wave periods), which corresponds to the conditions of a resonant $\pi/2$ pulse. The Rabi oscillation occurs at the slowest frequency because the effective z field seen by the spins is minimized in the RF frame. In contrast, further off resonance the oscillations become faster, due to the off-resonant field component along z and the amplitude is lower due to the smaller x/y component of the magnetization that is generated. All of these data can be fit by the single unknown parameter, B_x , and indeed do quite well as shown by the red curves in Fig. 11b where $B_x = 11.3 \mu\text{T}$. The nominal pulse length is therefore determined to be $\tau_\pi = 116.7 \mu\text{s}$ and the number of square pulses is $N = 7$, giving $\theta = 0.112$ or 4.6° . As a further result, we find the field-to-current ratio for the pulse coil: $\text{d}B_x/\text{d}I = 90 \mu\text{T A}^{-1}$.

In a similar experiment we measured Rabi curves for ^7Li in the solution of LiCl described previously (Section 4.1), obtaining a values of $\tau_\pi = 320 \mu\text{s}$ and $N = 7$ for a resonant $\pi/2$ pulse. These values are expected, since the resonance frequency (thus $1/\tau_\pi$) should scale by $\gamma_{\text{Li}}/\gamma_{\text{H}} \approx 0.389$ while N is independent of γ .

We conclude with a remark that the speed of the Rabi oscillation in the resonant square wave case is $2N\theta/(N\tau_\pi) \sim (2/\pi)|\gamma B_x|$ in angular units. The same result is found using the rotating frame method, where the oscillation speed is equal to the amplitude of the resonant field; the factor of $(2/\pi)$ comes from the $\exp[-i\pi(t/\tau_\pi)]$ Fourier component in the function that describes the resonant alternating field, $B_x(t) = (4/\pi)[\sin(\pi t/\tau_\pi) + (1/3)\sin(3\pi t/\tau_\pi) + (1/5)\sin(5\pi t/\tau_\pi) + \dots]$.

Other examples of indirect detection are inversion- or saturation-recovery measurement of the longitudinal relaxation time, T_1 , the Nuclear Overhauser Effect (NOE) and Chemical Exchange Saturation Transfer (CEST). These measurements could be implemented in a way similar to the above, using multiple event sequence files and the list structures to record spectra with multiple different values of inter-pulse delays or other parameters.

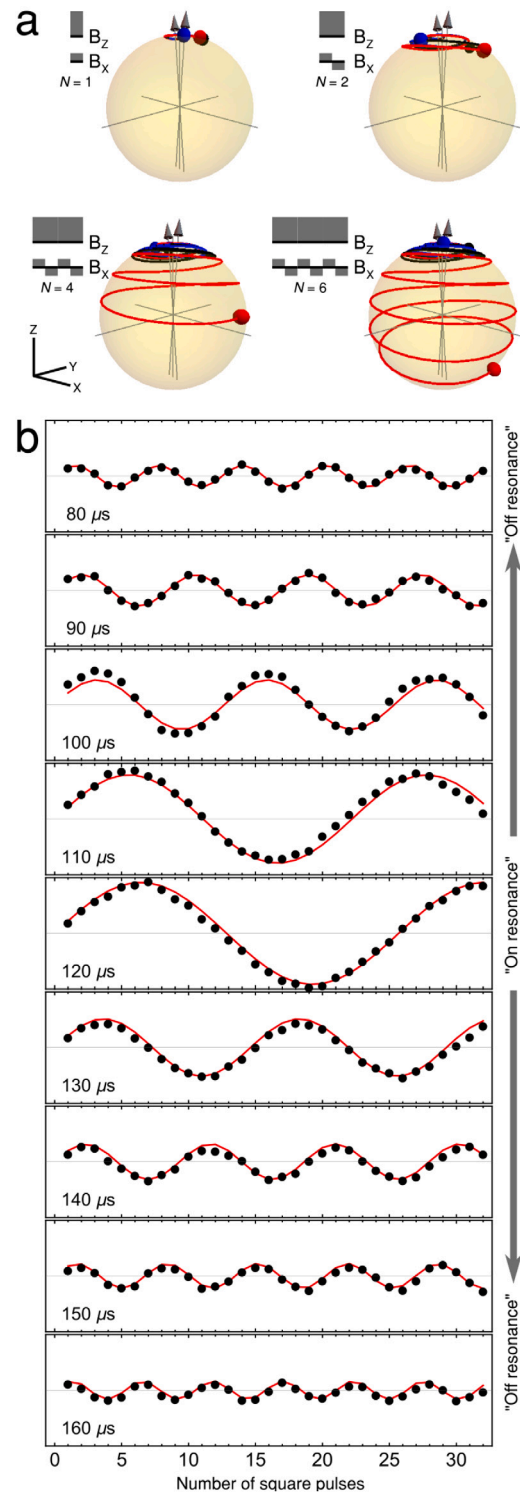


Fig. 11. Resonant excitation with NMRduino. (a) Bloch sphere trajectory of z magnetization during square-wave pulses of different period 2τ and duration $2N\tau$. The on resonance case (red, $\tau = \tau_\pi \equiv \pi/[\gamma(B_x^2 + B_z^2)^{1/2}]$, for $B_x/B_z = \tan(10^\circ)$) shows linear accumulation of nutation angle with number of pulses N , whereas off-resonance the accumulation is destructive (black, $\tau = 0.8\tau_\pi$ and blue, $\tau = 1.2\tau_\pi$); (b) experimental curves of ^1H NMR signal amplitude vs. τ and N for $^1\text{H}_2\text{O}$ in $B_z = 100 \mu\text{T}$ (black circles). Red curves represent a global fit of the x component from (a) to the data, using the single parameter $B_x = 11.3 \mu\text{T}$ ($B_x/B_z = \tan(6.4^\circ)$; $\tau_\pi = 116.7 \mu\text{s}$).

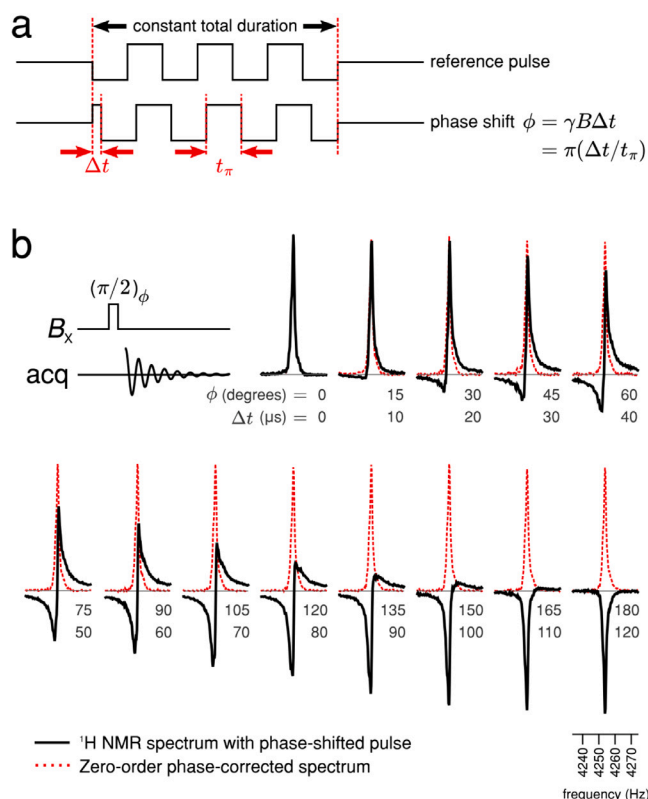


Fig. 12. Pulse phase shifting using NMRduino: (a) phase-shift concept for resonant square-wave pulses; (b) experimental data for ^1H in H_2O at $100\ \mu\text{T}$ ($|\gamma_{\text{H}} B / (2\pi)| \approx 4250\ \text{Hz}$), single scans, prepolarization at 1 T. Here, ^1H precession signals are recorded a constant time (10 ms) after applying a square-wave pulse of nominal flip angle $\pi/2$, $N = 7$, for different pulse time shifts, Δt , in increments of $10\ \mu\text{s}$. The pulse phase $\phi \approx 8500\pi\Delta t$ rad produces an equal phase shift in the raw signal (black curves), which is confirmed by phase correction of the spectra (dotted red curves) through taking the real part of raw signal after multiplication by $\exp(2\pi i\phi)$.

4.3. Phase-shifted pulses

In the typical NMR setup, an RF coil transmits the oscillating magnetic field along only one axis – again for simplicity let us say the x axis – and by changing the frequency this allows one to rotate spins about any axis in the xz plane. By shifting the RF phase, one can also reorient the pulse's effective axis in the xy plane. As an example, the resonant square-wave pulse introduced in Section 4.2 can be phase shifted as demonstrated in Fig. 12a. A phase shift of ϕ radians is introduced by time-translation of the waveform by $\phi/(2\pi)$ times its period, and this corresponds to an effective z rotation of the nutation axis by ϕ in the Bloch sphere (RF frame). A phase shift of $+90^\circ$ ($\phi = +\pi/2$) therefore moves the nutation axis from the x axis to the y axis.

The availability of high-resolution high-speed DACs means that a modern NMR spectrometer will use direct digital synthesis (DDS) to generate RF waveforms and their associated phase shifts. A pulse waveform is represented digitally as a circular array of amplitudes for discrete time points during a single cycle of the RF carrier sine wave, a.k.a. a lookup table. The spectrometer increments through the table at the set rate, reading off amplitude values and sending them to the DAC. This process is akin to how NMRduino adjusts the amplitudes of the A0 and A1 channel outputs, though on the modern spectrometer it is done on a much faster time scale and using dedicated DDS hardware. However, to introduce a phase shift in the waveform the process is similar: one simply jumps ahead in the lookup table. For a table of length n , an increment of k points leads to a phase shift of $2\pi k/n$ radians. A 90° or 270° phase shift in the sine wave, respectively, requires the DDS to jump by one quarter or three quarters of the length

of the lookup table. On NMRduino (for now) the equivalent of the lookup table is the event sequence file, where phase shifts are specified by changing the time stamp of DAC switching events. In both cases the update rate of the controller determines the resolution of the phase shift: in the lookup table it is $2\pi/n$ while for NMRduino it is $1\ \mu\text{s}$ times the Larmor frequency (in rad s^{-1}). At $100\ \mu\text{T}$ or a frequency of $4\ \text{kHz}$ for ^1H , the phase resolution is approximately 1.5° as can be confirmed experimentally, see Fig. 12b.

Pulse phase shifting is especially important for NMRduino users who wish to perform multi-pulse NMR experiments with a single transmit coil. These might require the magnetization to be perturbed along a specific axis, a common case of which is within the spin echo, or cycled in different scans of the experiment to lead to constructive or destructive interference between coherence transfer pathways. Another application of phase cycling is time-proportionate phase incrementation (TPPI), used to discriminate the direction of precession in two-dimensional NMR spectroscopy.

However, one is not always restricted to a single coil axis. For example, below around 10 mT a set of three DC pulse coils can be arranged around the sample to perform rotations about arbitrary axes in space [84,85], without phase shifting. These can also be error compensated [48,86].

4.4. Spin echoes

Spin echoes are a fundamental technique used across NMR relaxometry, spectroscopy and MRI where precessing magnetization must be refocused after being dephased, leading to a resurgence or “echo” of the signal. These can be used to correct for inhomogeneity in the magnetic field that might otherwise cause signal loss, or to accumulate evolution under other interactions, e.g., scalar couplings.

Any reputable NMR or MRI textbook will explain how the echo phenomenon arises. However, for clarity, we briefly describe the principle again here. Starting from nuclear magnetization parallel to the axis of the main magnetic field, a pulse is applied to flip the magnetization into the orthogonal plane (nominally, of flip angle $\pi/2$). After the pulse the magnetization precesses around the main field as described before, but gradually loses coherence due to magnetic gradients: fast-precessing spins in the sample race ahead while slow-precessing ones fall behind. After a delay time τ , before the signal has completely lost coherence, a second pulse of angle π is applied. This inverts the spins such that the faster precessing spins are now lagging behind and the slower ones are ahead, so that after a further time delay τ all spins meet back together in phase.

An extension of the basic echo method is the Carr–Purcell–Meiboom–Gill (CPMG) sequence [87,88]. The CPMG consists of not one, but a series of equidistant π pulses to generate multiple echoes in succession, which can each be detected and measured. As an experimental example, we refer to Fig. 13 which shows a series of multiple-echo signals acquired for the 1 mL water sample at $100\ \mu\text{T}$. The π pulses were generated as described in Section 4.2 using $117\ \mu\text{s}$ as the square pulse length on channel A0. After each π pulse the magnetometer signal was acquired (2048 data points, 20,000 samples/s) and in each of these data windows one can see the temporary revivals of the signal caused by the echo. The three sets of signals in Fig. 13 correspond to different gradients, dB_z/dz , that were deliberately introduced into the background field through a shim coil to accelerate coherence dephasing. These ranged from $10\ \text{nT cm}^{-1}$ (top) to $70\ \text{nT cm}^{-1}$ (center) to $100\ \text{nT cm}^{-1}$ (bottom). Generally speaking, the overall decay envelope was insensitive to these gradients and consequently the CPMG technique allowed the true transverse decay time constant of the sample to be measured, $T_2 \sim 1.4\ \text{s}$. In comparison the free-decay signals without applied π pulses is much faster, and the rate constant $1/T_2^*$ increases approximately linearly with the gradient strength (see below each echo-train signal in Fig. 13). To record these signals without echoes, an almost-identical event sequence file was used, where the output of the

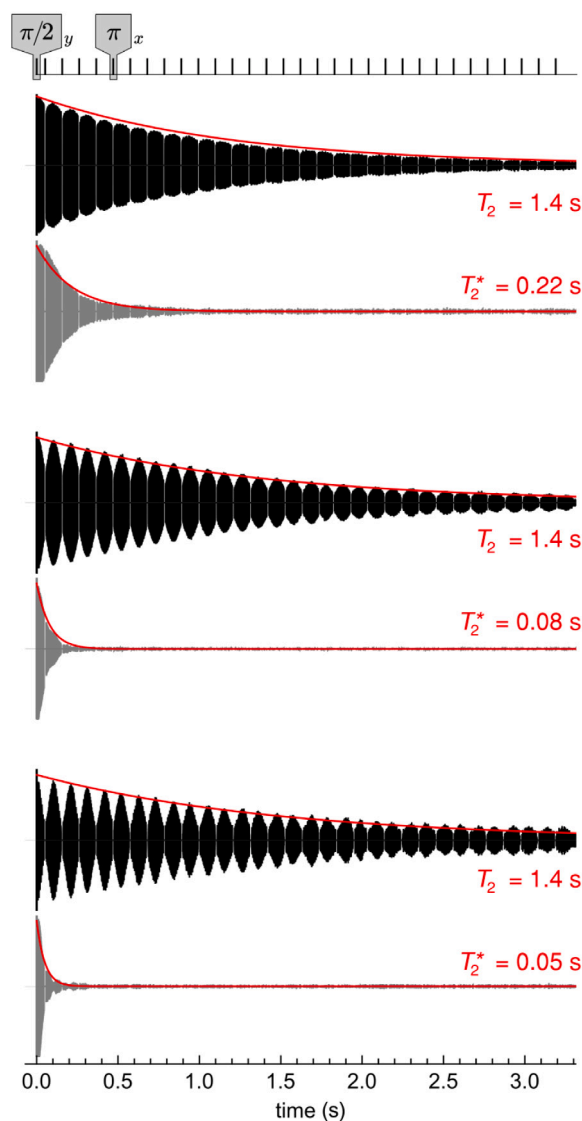


Fig. 13. The use of NMRduino to record ^1H precession signals (4258 Hz, $100\ \mu\text{T}$) during Carr-Purcell-Meiboom-Gill spin echo trains. Multiple-echo (black) and free-precession signals (gray) for 1 mL of 0.8-T-prepolarized H_2O are plotted in pairs for three values of an applied field gradient along the z axis: $10\ \text{nT cm}^{-1}$ (top), $70\ \text{nT cm}^{-1}$ (center) and $100\ \text{nT cm}^{-1}$ (bottom). The red curves denote fits of the experimental data to monoexponential decay functions with time constant T_2 or T_2^* . Each trace is the sum of 50 scan averages, detected via an atomic magnetometer [47]. All signals were band-pass filtered with a Hamming window function, passband 4200–4300 Hz.

A0 DAC was set to zero after the first $\pi/2$ pulse. Other events and timings in the sequence were unchanged.

There are many applications and extensions of spin echoes, which we encourage NMRduino users to explore. The measurement of T_2 decay constants is highly relevant in the study of molecular dynamics. Windowed-acquisition in between echo pulses is also closely related to the techniques used in MRI [88]. After an initial RF pulse is applied, a series of pulses refocus the nuclear magnetization depending on spatial position, leading to the creation of spin echoes that form the basis of the image. Future versions of NMRduino may indeed support tools for low-field MRI.

5. Advanced examples

The following section demonstrates NMR experiments using hetero- or multi-nuclear systems. Beginners may wish to save this section for future reading.

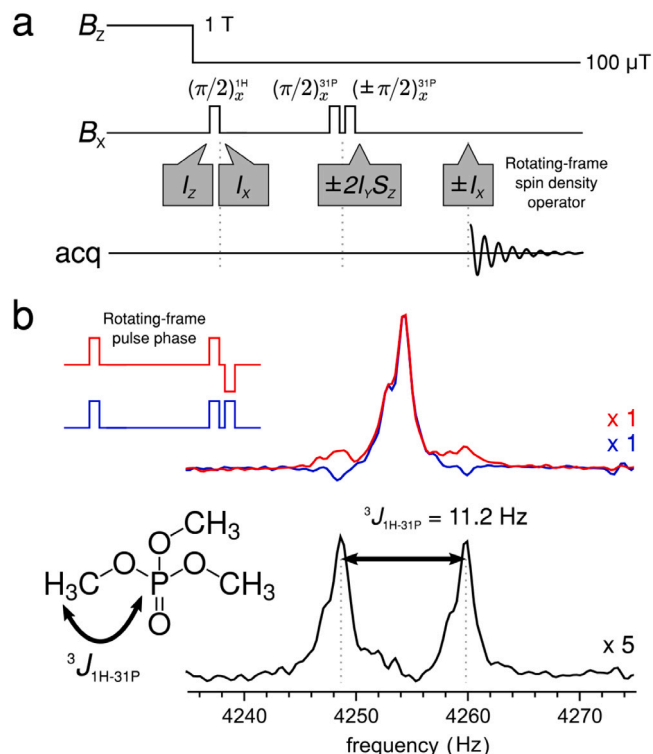


Fig. 14. Multinuclear coherence selection using NMRduino: (a) pulse sequence for heteronuclear coherence filtering. Time delays between the first and second pulse, and between the third pulse and acquisition is nominally set to $\tau_{\text{het}} = 1/|2J_{IS}|$; (b) ^1H -detected NMR spectra of 2 M $(\text{CH}_3\text{O})_3\text{PO}$ in H_2O using the pulse sequences from (a) with $\tau_{\text{het}} = 42\ \text{ms}$. The overlaid signals each correspond to the different phase of the final $\pi/2$ pulse ('+' = red curve, '-' = blue curve). The subtraction of the two signals is shown below (black curve).

5.1. Coherence filtering

When the NMR sample comprises a mixture of chemicals it is common to use pulse sequences that selectively observe the signals from desired molecules and suppress the signals from the rest. This scenario includes ^1H -detected NMR of chemicals with couplings to ^{13}C spins at natural isotopic abundance, such as in methanol (a mixture of $^{12}\text{CH}_3\text{OH}$ and $^{13}\text{CH}_3\text{OH}$) or acetic acid ($^{12}\text{CH}_3^{12}\text{COOH}$ plus $^{12}\text{CH}_3^{13}\text{COOH}$ and $^{13}\text{CH}_3^{12}\text{COOH}$, and to a much lesser extent $^{13}\text{CH}_3^{13}\text{COOH}$) where it is desirable to attenuate the strong signal of the all- ^{12}C isotopologs in order to observe the ^{13}C -satellite signals and their associated heteronuclear J couplings. Various sequences exist for this task [45], namely BIRD (Bilinear Rotation Decoupling) filtering, TANGO (Testing for Adjacent Nuclei with a Gyration Operator) and heteronuclear double quantum filtering, which forms the basis of the famous INADEQUATE experiment (1D-Incredible Natural Abundance Double-Quantum Transfer Experiment).

The common feature of these pulse sequences is that they contain time delays for free evolution under the heteronuclear scalar coupling. Consider the three-pulse sequence given in Fig. 14a. The initial pulse of ^1H flip angle 90° excites in-phase magnetization along the x direction and a subsequent delay of $\tau_{\text{het}} = 1/|2J|$ allows these to evolve completely into heteronuclear two-spin order (i.e., $2I_xS_z$ for a spin pair $I = 1/2$ [^1H] and $S = 1/2$) as labeled by the product operators. The following two 90° pulses – this time applied to the heteronucleus – are intended to selectively manipulate the sign of the bilinear coherence operator. Where the '+' phase is used, the sign is inverted, while for the '-' combination the two pulses cancel each other out, and the operator sign is unchanged. In contrast, I -spin operators not coupled to the heteronucleus remain unchanged in sign. Subtraction of the

NMR signals from alternate scans of '+' and '-' pulse phase results in accumulation of the signals from the heteronuclear couplings of interest, while signals from uncoupled spins cancel.

An experimental demonstration of the heteronuclear coherence filter using NMRduino is shown in Fig. 14b. The sample used is a dilute solution of trimethylphosphate in water (20% vol. $(\text{CH}_3\text{O})_3\text{PO}$ in H_2O ; three-bond scalar coupling $^3J_{\text{H-}^{31}\text{P}} \approx 11$ Hz) and the signal is measured in a poorly shimmed 100 μT field, or at around 4.26 kHz ^1H Larmor frequency, again with an atomic magnetometer [47]. The experimental apparatus overall follows the same setup as Fig. 10a, but with pulses of scaled duration compared to the ^1H case for resonant excitation on ^{31}P ($\tau_\pi = \gamma_{\text{H}}/\gamma_{\text{P}} \times (117 \mu\text{s}) \approx 300 \mu\text{s}$). Two event sequence files are used, one each for the '+' and '-' versions of the pulse sequence in Fig. 14a.

The two colored spectra in Fig. 14b correspond to the '+' and '-' pulse sequences and agree with the prediction that the $-\text{OCH}_3$ multiplet should reverse sign relative to the central H_2O line. The difference of these two spectra is plotted at the bottom of the figure, and clearly shows the resonances only due to the heteronuclear coupled ^1H spins.

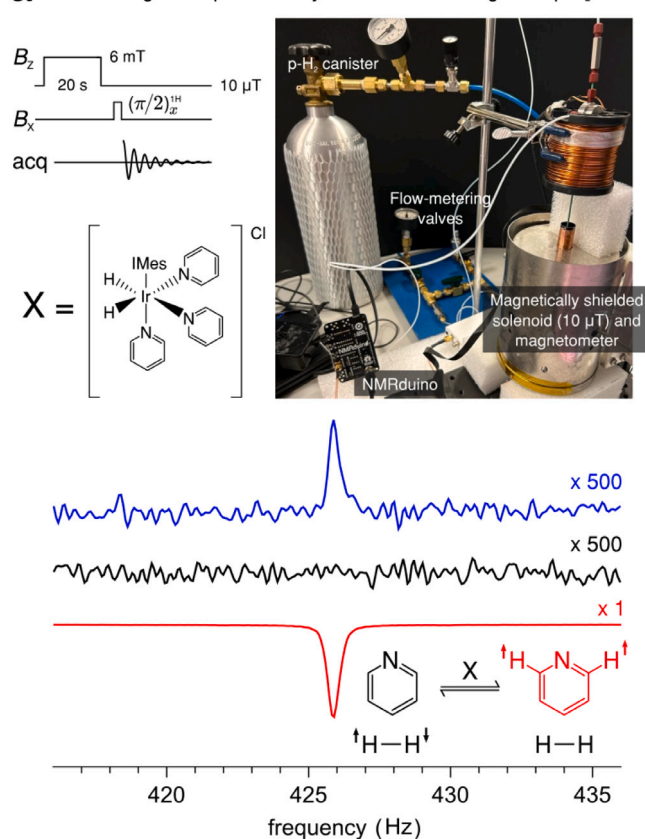
5.2. Spin order transfer and hyperpolarization

The low-field regime can also be exploited for transferring magnetic order between spin species, an outstanding and useful case of which is to tap the nuclear spin ordering of para-enriched hydrogen ($p\text{-H}_2$) to yield signals of intensity several thousands of times stronger than via Zeeman polarization [89,90]. A typical $p\text{-H}_2$ -enhanced experiment involves applying a sequence of μT -mT fields during a chemical reaction to transfer the ^1H spin-singlet order of $p\text{-H}_2$ into ^1H or ^{13}C magnetization via heteronuclear couplings. The aim is to ensure a smooth correlation of spin states between $p\text{-H}_2$ and resulting chemical products so that the highly ordered starting spin state is carried over to new molecules — and that these become *hyperpolarized*, with polarization levels on the order of 10% to 100% [91].

Signal Amplification By Reversible Exchange (SABRE) is a specific method where hyperpolarization takes place during dynamic chemical association-dissociation exchange of (i) $p\text{-H}_2$, (ii) the target to be hyperpolarized and (iii) a catalyst species that binds the molecules (i)+(ii) together. The prototypical example of SABRE is to polarize ^1H spins in the heterocyclic compound pyridine ($\text{C}_5\text{H}_5\text{N}$) in fields of around 6 mT [92], which we reproduce and observe using NMRduino in Fig. 15a. The 6 mT polarization field along z is supplied by a coil of wire connected to the parallel high-current channel D2, on top of a 10 μT detection field supplied by the magnetically shielded solenoid. Signal detection at 10 μT or ^1H Larmor frequency 426 Hz is performed using a commercial atomic magnetometer (QuSpin Inc. QZFM2, sensitive axis along x) connected to the NMRduino's analog input, following a DC $\pi/2$ pulse on the ^1H spins (125 μT for 47 μs). The pulse is supplied by a third coil which makes a field along the x direction when energized by the output channel D0. The spectra in Fig. 15a are single-scan recordings for three separate samples, all at room temperature, as follows: (1. top, in blue) 1 mL distilled water, after thermal/Zeman prepolarization at 6 mT; (2. center, in black) 0.1 M pyridine in 1 mL perdeuterated methanol (CD_3OD), hyperpolarized at 6 mT in the presence of the iridium-Imes catalyst, 'X'; (3. bottom, in red) the same as 2., with $p\text{-H}_2$ bubbled at 5 bar through the solution at a rate of $\sim 1 \text{ mL s}^{-1}$. The ^1H NMR signal in 3. is around 500 times stronger than that in 1., despite a 2250-fold lower concentration of ^1H . We see as a result that $p\text{-H}_2$ enlarges the (mean) ^1H polarization by a factor of 1.1×10^5 . There are many extensions of this method, such as multinuclear detection, quantitative testing, and continuous monitoring of hyperpolarization procedures that we leave the reader to investigate using NMRduino.

A similar low-field method is called SABRE-in SHield Enables Alignment Transfer to Heteronuclei (SABRE-SHEATH [93,94]). Here the dynamic system is held at a lower field of around $\sim 0.5 \mu\text{T}$ during bubbling, which serves to transfer ^1H spin-singlet order of $p\text{-H}_2$ into

a SABRE: Signal Amplification By Reversible Exchange with $p\text{-H}_2$



b SABRE-SHEATH = SHield Enables Alignment Transfer to Heteronuclei

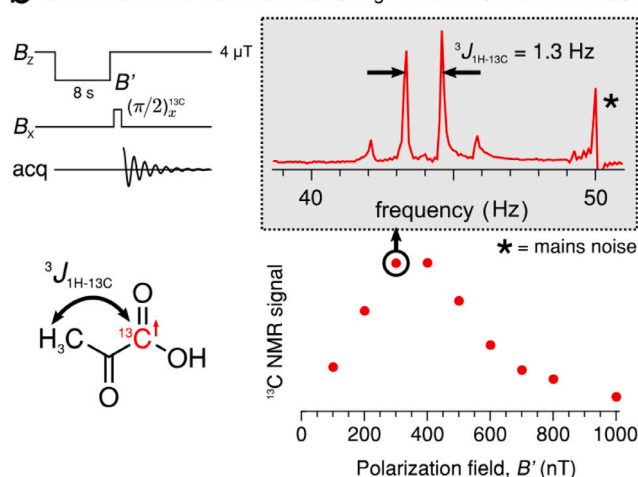


Fig. 15. Production and observation of hyperpolarized NMR signals using NMRduino: (a) 6 mT SABRE enhancement of ^1H . Colored spectra correspond to: (blue) a 1 mL reference sample of distilled $^1\text{H}_2\text{O}$, thermally polarized without $p\text{-H}_2$; (black) a 1 mL solution of 0.1 M pyridine in CD_3OD , plus 10 mM of catalyst 'X'; (red) the same pyridine solution during $p\text{-H}_2$ bubbling (1 mL s^{-1} , 5 bar); (b) SABRE-SHEATH ^{13}C polarization dependence at sub- μT fields in a solution of 30 mM $[1\text{-}^{13}\text{C}]\text{-pyruvate}$, 40 mM DMSO and 6 mM $\text{Ir}(\text{Imes})(\text{COD})\text{Cl}$ in CD_3OD at 10°C under $p\text{-H}_2$ bubbling (1 mL s^{-1} , 5 bar). The displayed spectrum corresponds to the ^{13}C resonance at 4 μT . The overall $^{13}\text{C}_1$ polarization was around 1%–2%.

^1H , ^{15}N or ^{13}C magnetization on the target, depending on which of these spins are present and the sizes of the scalar (J) couplings involved.

The magnetic field required for SABRE-SHEATH is chosen to coincide with an avoided level crossing within the Zero- to Ultra-Low-Field

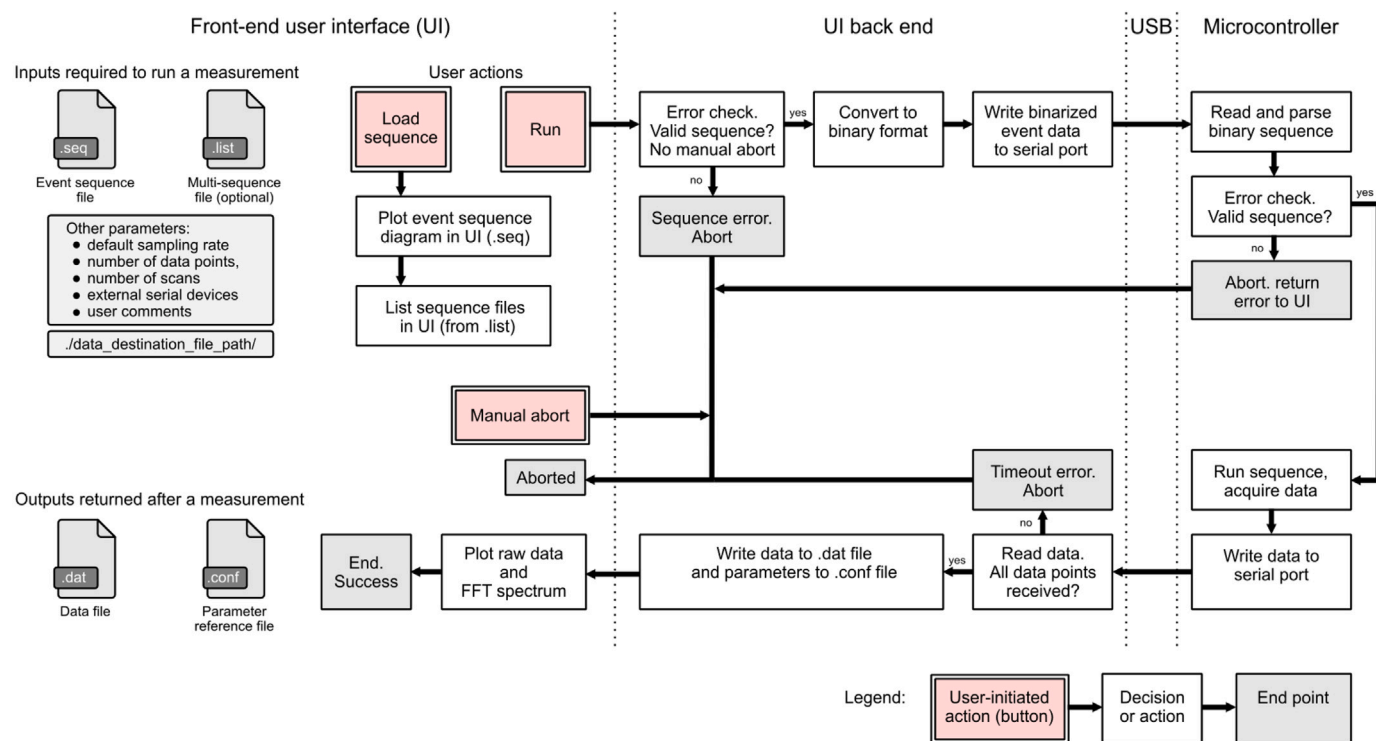


Fig. 16. Detailed software layer structure of NMRduino. The front end of the UI principally serves to execute event sequences through the 'Run' command and graphically visualize the data, as well as the sequences, on screen. The flow chart shows how the UI's back end handles error checking and connections with the lower-level MCU software layer.

(ZULF) regime on the order of $B \approx 2\pi J_{IS}/|\gamma_I - \gamma_S|$, where J_{IS} is the heteronuclear coupling and $I = {}^1\text{H}$. The specific field condition is quite narrow, typically within the range of 100 nT; the name SHEATH is therefore derived from the necessity of using a magnetic shield to access these fields effectively, allowing for the passive screening of the Earth's magnetic field. However, generating a low magnetic field condition is relatively straightforward, especially with the capabilities of NMRduino. One can connect the high-current outputs A0 or A1 to a solenoid coil as described before. Alternatively, given the 25 mA maximum current rating of the MCP4822 DAC, and the favorable field-to-current ratio of the solenoid ($7.5 \mu\text{T mA}^{-1}$), the fields can be supplied by connecting the solenoid terminals to the DAC output directly, bypassing the need for amplifier-H-bridge intermediaries.

An illustrative case of NMRduino applied to SABRE-SHEATH is given in Fig. 15b. The experimental setup is similar to that shown in the photograph in Fig. 15a where the magnetometer and the sample are both inside the magnetic shield. Pulse-acquire scans may be measured for a series of different DAC output values to observe the hyperpolarization yield at various magnetic field strengths. One way to achieve this is by programming different DAC values in different sequences, and running these in series using a '.list' file. This rapidly produces a curve depicting the field-dependent efficiency of polarization, as shown in the figure where ${}^{13}\text{C}$ precession signals are measured at 4 μT , or 533 μA solenoid current. A suitable shunt resistor is connected in series with the solenoid to adjust the dynamic range; a resistance of 10 k Ω means that the 10 V maximum supply voltage translates to a maximum output current of 1 mA, and furthermore that the DAC output resolution translates to a field precision of $7.5/2^{12} \mu\text{T} = 1.8 \text{ nT}$.

The procedure shown applies to polarization of ${}^{13}\text{C}$ on the target compound [1- ${}^{13}\text{C}$]-pyruvate ($\text{CH}_3\text{CO}^{13}\text{COO}^-$, 30 mM in D_2O , 10 $^\circ\text{C}$), chosen because the hyperpolarized form of the ion is an important spin tracer applied widely in metabolic MRI [95,96]. The best DC field condition for ${}^{13}\text{C}$ polarization appears to be between 300 and 400 nT for pyruvate [53]. This range is in agreement with previous

literature [97]. In those cases, the sample is manually transferred outside of the magnetic shield for NMR measurement in a separate and dedicated high field ($\sim 1 \text{ T}$) instrument. The shuttle-less polarization and magnetometer detection approach offered by NMRduino is highly convenient and furthermore highly cost effective. As a result we expect NMRduino to be used widely in polarimetry studies [98].

A minor experimental detail of both examples is that the $p\text{-H}_2$ gas is continuously bubbled through the solution during both the polarization and acquisition phases of the experiment. Due to the low field strength, any magnetic susceptibility changes caused by the bubbling are too weak to cause inhomogeneous broadening, and the resonance line remains sharp with a line width below 1 Hz. However, if desired, a stopped-flow version of the experiments could instead be envisioned where gas-metering valve actuators are controlled by commands from the ".seq" files, typically via the D0-D2 lines.

An additional characteristic of SABRE-SHEATH is that the intramolecular Zeeman splittings and spin-spin couplings are of comparable magnitude to the rates of intermolecular chemical exchange. Improvements in the efficiency of polarization transfer can be achieved compared to the static level-anticrossing condition by solving these more complex dynamics to obtain modulated magnetic fields, such as duty-switched SABRE [99] and the Multi-Axis Computer-aided HETeronuclear Transfer Enhancement, or MACHETE-SABRE sequences [100–102]. We also expect NMRduino to be useful here as an inexpensive, versatile programming device to test these sequences experimentally [103]. Finally, the analog high-current channels A0/A1 could also be used to perform sweeps across the heteronuclear avoided crossings. One example is the [1- ${}^{13}\text{C}$]-fumarate system, which can be hyperpolarized in the ${}^1\text{H}$ -singlet form by trans addition of $p\text{-H}_2$ to the triple carbon-carbon bond of [1- ${}^{13}\text{C}$]-acetylenedicarboxylic acid. The singlet order is subsequently converted rapidly ($<1 \text{ s}$) into either ${}^1\text{H}$ or ${}^{13}\text{C}$ magnetization with efficiency in practice reaching above 50% upon adiabatically changing the field regime between singlet-triplet eigenstates and high-field (Zeeman) eigenstates [104–106]. Like MACHETE, shaped profiles of current vs. time could be programmed into the event sequence file and combined with acquisition.

6. Conclusions and outlook

In this paper we have introduced a versatile and accessible low-frequency NMR platform: NMRduino, so-called after its creation from tools and know-how leveraged from the popular Arduino open-source-electronics ecosystem. The experimental demonstrations provided show feasibility and effectiveness in multinuclear NMR spectroscopy, spin relaxation measurements and hyperpolarization procedures with appeal – we hope just like Arduino – to researchers and enthusiasts alike.

A key strength of NMRduino even in the present release version is a potential for customization and expansion. Given its open-source nature [34], users who wish to adapt the platform to suit their specific needs may easily do so by integrating additional sensor/probe or coil hardware, or by implementing advanced signal processing algorithms or autonomous control features in the software and firmware. The overall structure of NMRduino gives is also resistant to major upgrades in third-party products such as the MCU. In fact, we have already seen this in practice — we upgraded NMRduino to run on the Teensy 4.1 MCU after its predecessor, the Teensy 3.5, ceased production and support in early 2023. The two MCUs were mostly pin-for-pin compatible and therefore only minor changes to the PCB layout were needed, while the only software changes were the MCU-specific instructions to initialize and run the hardware-interrupt timer in the firmware. We consider this to be very light work compared to the process of rewriting specific software for FPGA-based spectrometers [42].

Another feature of NMRduino is that it exploits generic, low-cost and modular hardware. Chips such as H bridges are bought ready to use, but also are cheap enough to throw away, opening up the interesting possibility of NMRduino being used as a “disposable” NMR spectrometer. As a result we predict that NMRduino may contribute a new and vast array of possibilities for NMR in interdisciplinary research, in resource-constrained environments and in fieldwork. We therefore encourage researchers, educators and enthusiasts to explore NMRduino to advance their own investigations, and share their developments and new contributions back with the community. We are optimistic that this process will play a pivotal role in pushing the boundaries of NMR user skills to catalyze new scientific breakthroughs.

CRedit authorship contribution statement

Michael C.D. Tayler: Writing – review & editing, Writing – original draft, Supervision, Software, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.
Sven Bodenstedt: Writing – review & editing, Software, Methodology, Investigation, Data curation, Conceptualization.

Declaration of competing interest

The authors declare no competing interests.

Data availability

All data appearing in the published article will be freely available via the OPENAIRE data base, hosted on Zenodo: URL <https://doi.org/10.5281/zenodo.10895054>.

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Appendix

A.1. Event sequence handling

As shown in Fig. 5, the procedure to commence the NMR experiment starts when a user clicks the “Run” button of the NMRduino UI and this proceeds as follows. First the UI checks whether a valid pulse sequence path has been given, and if so, loads the file data into its memory. It then tests for possible basic errors in the pulse sequence before proceeding further, by asking for example: (1) is the number of events in the header file unequal the number of event rows? (2) does the event sequence exceed safe power limits on the high-current digital and analog output channels, or leave these outputs enabled at the end of the event sequence? (3) Are there timing conflicts between any two pulse or acquire events? If the response to any of the test questions is a ‘yes’ then the event sequence is determined to be invalid, and the Run command aborts.

If the responses are all ‘no’ then the pulse sequence proceeds to be transferred to the microcontroller. The first step of this process is to convert each row of the event matrix into a binary string comprising the event timestamp (32 bits in length) followed by the event data (another 32 bits in length), which corresponds to the digital GPIO for H-bridge control, acquisition, or the amplitude of the DAC output depending on a specified control word that is included in the string. Each event is therefore expressed as an 8-byte binary string: four bytes for the timestamp plus another four for the event data. The maximum of 20,000 events is determined from the spare memory ($20,000 \times 4$ bytes = 80 kB) available for storage on the microcontroller’s RAM partition. A conversion for the illustrative event sequence given in Fig. 6 is provided in Fig. 17, using the GPIO mapping given in Table 1.

Certain events are, at the moment, forbidden in NMRduino pulse programs. One of these, due to timing restrictions imposed by the MCU firmware, is that DAC voltage switching, data point acquisition, and GPIO switching must be given as separate events; requesting any of these together in a single event will result in an error. A further constraint is that a DAC value set to zero ($A0_dac = 0$) is ignored, in order to simplify programming and keep number of events low. As a workaround, zeroing of the A0/A1 channel outputs should be done by switching the GPIO. Both of these limitations could be addressed in future by more complex programming rules such as assigning types of event to a hierarchy, so that they can be chosen according to order of priority.

A.2. Event sequence upload

After being successfully checked and converted to binary form, the event sequence is written to the MCU’s USB COM port at a data rate of around 1 Mbits/s (~15,000 events/s). A handshake-type protocol ensures correct upload of the event sequence and identify any missed bytes that potentially can translate into hardware-damaging errors in the output at the MCU end. Failure to complete all steps of the handshaking procedure causes the “Run” command of the UI and the event upload to abort. These steps, in order, are as follows:

- The UI first writes a short (2-byte) string of known data to let the MCU know that it intends to upload an event sequence, followed by another two bytes whose value N_{events} is equal to the total number of events. It is assumed that the MCU is in an idle state, waiting for serial data to be received before performing any further action.

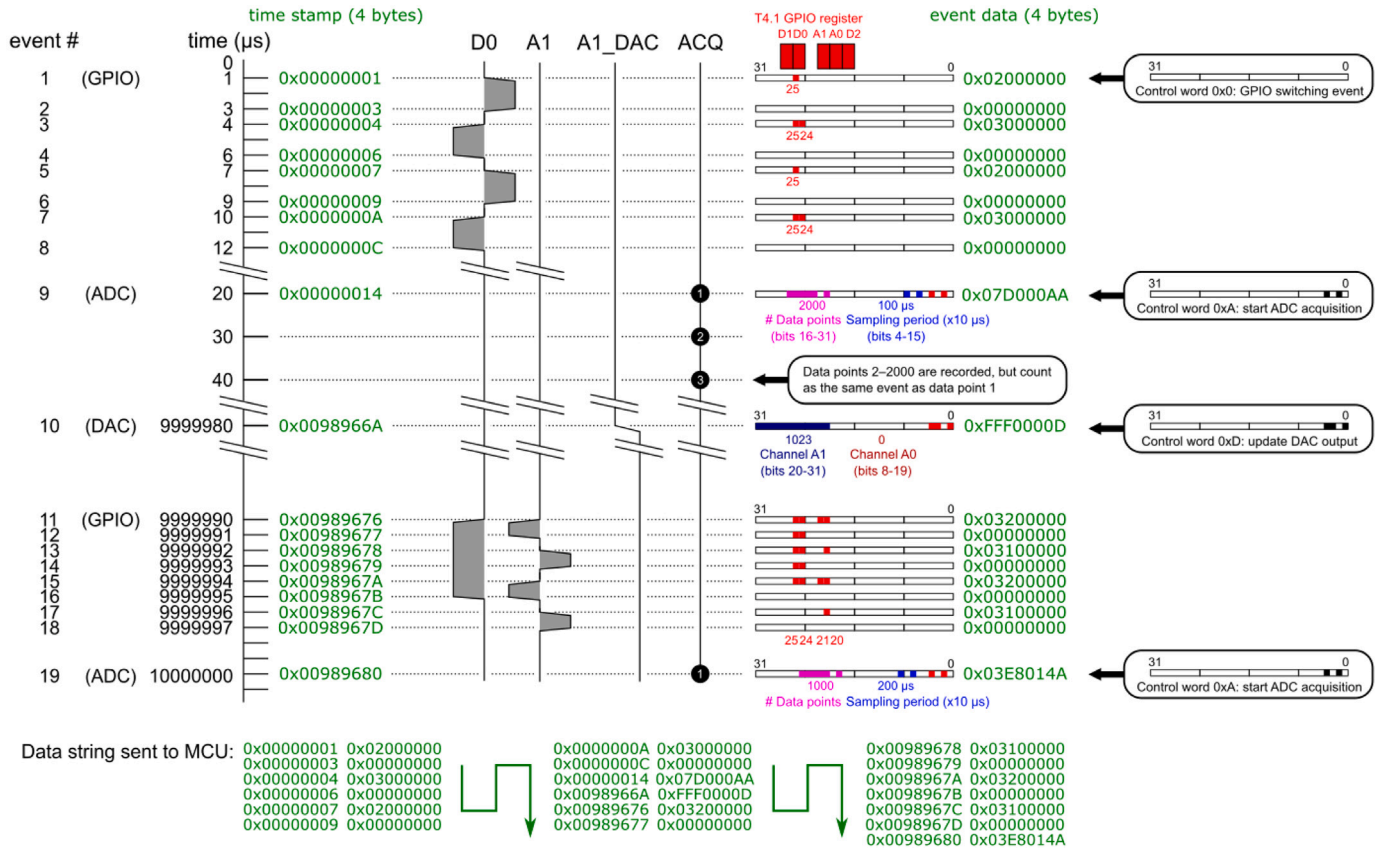


Fig. 17. Mapping of the example .seq file data from Fig. 6 into binary format for USB upload to the MCU, given the GPIO encoding summarized in Table 1 plus additional syntax rules defined here. The entire pulse sequence of duration 10 s has been reduced to 19 events, totaling 152 bytes. For convenience, the sequence data (in green) are expressed in hexadecimal.

Table 1

Mapping between NMRduino output channels and the corresponding control bits in the 32-bit GPIO register of the Teensy 4.1 MCU. The bit function is denoted either as “a” for amplitude control (0 = output disabled, 1 = output enabled) or “d” for direction control of the bipolar current output (0 = positive, 1 = negative).

NMRduino channel	GPIO function	IMXRT1060 GPIO register name	Teensy 4.1 pin
D0	a	GPIO6_25	23
	d	GPIO6_24	22
D1	a	GPIO6_27	21
	d	GPIO6_26	20
D2	a	GPIO6_16	19
	d	GPIO6_17	18
A0	d	GPIO6_19	15
	a	GPIO6_18	14
A1	d	GPIO6_21	41
	a	GPIO6_20	40
Unused	–	GPIO6_29	39
	–	GPIO6_28	38
	–	GPIO6_22	17
	–	GPIO6_23	16

- When these first serial data enter into the USB receive buffer of the MCU, a change of state is triggered. The data are read, and the MCU transmits back the first 2-byte string to the computer to confirm to the UI it is ready to receive the event sequence. The second two bytes (number of events) are written into the MCU memory.
- The UI proceeds to stream the binarized event data (such as those given in Fig. 17) over the COM port.

- The MCU reads the incoming stream and, by knowing the structure of the binarized sequence, maps the data onto a set of 32-bit unsigned integer arrays: one for the event time stamp list, two others for the GPIO “set” and “clear” registers and finally one for the A0-A1 DAC values. Parallel elements in these arrays correspond to the same event, namely that the n^{th} element of the GPIO register array (etc.) gives the new GPIO port status at the n^{th} timestamp.

During the transfer of sequence data over the COM port the MCU counts the number of events received. Upon counting to the expected number of events sent at the beginning of the handshake, any remaining data bytes in the serial buffer are read by the MCU and ignored. The COM port buffer is then flushed to conclude a successful upload of the event sequence. However, if the expected number of events is not reached after a reasonable timeout delay (100 ms), it is assumed that data bytes have been lost. An error message is sent back to the UI and the sequence is again aborted.

A.3. Timed event execution by the MCU

Upon successful upload of the event sequence (or after this step is skipped, as in signal averaging), the MCU proceeds to move into a state where the sequence is actually executed. Here the firmware initializes a feature called an *interrupt timer* for timing control, which is then attached to a function called an *interrupt service routine* (ISR). Without going into too much detail, the timer is a back-end function of the MCU that increments an integer count value by 1 every timer clock cycle, which is always some multiple of the CPU clock period. Upon the value of the counter becoming equal to some predetermined integer, often termed the *compare match value*, the timer triggers an interrupt

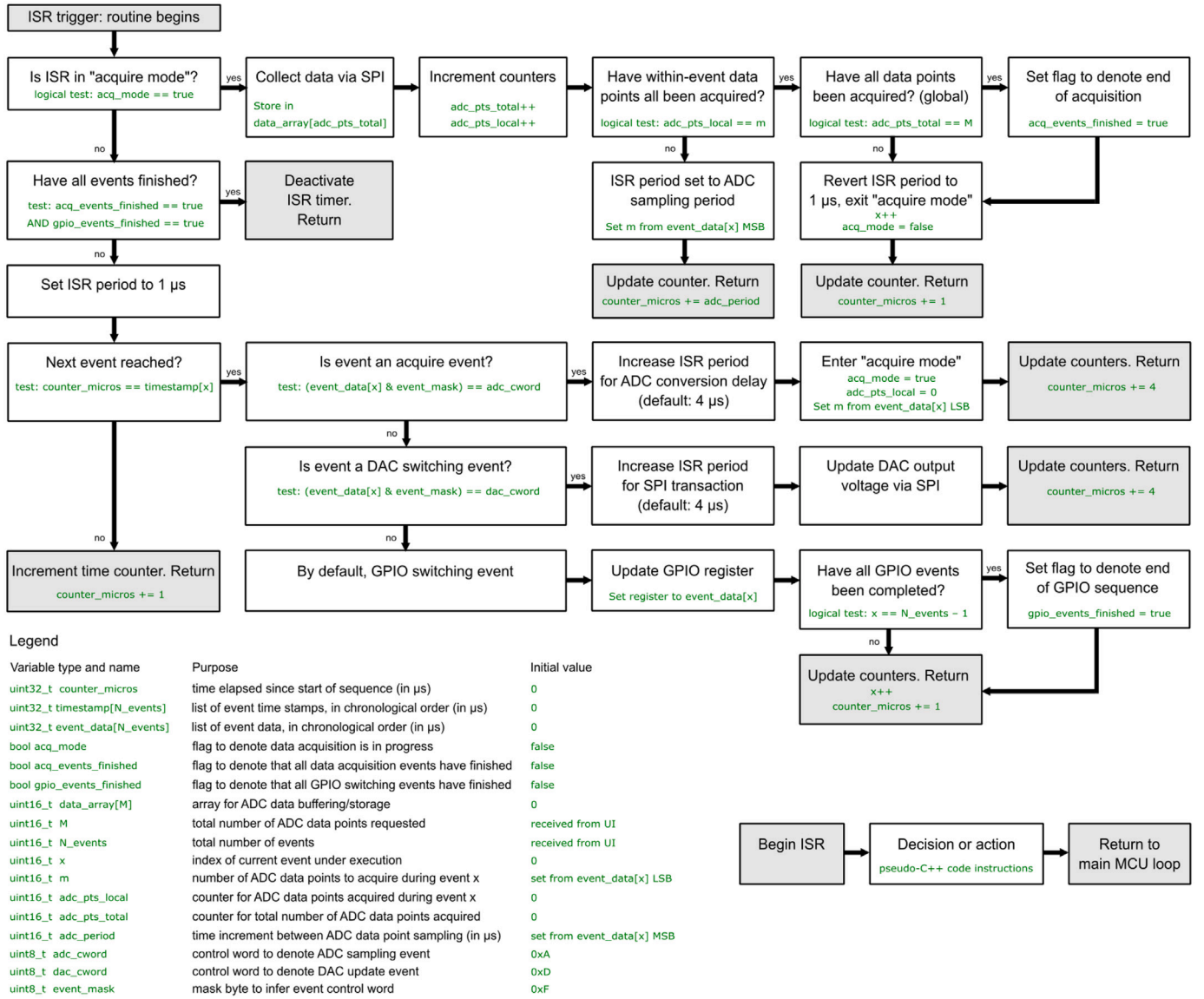


Fig. 18. A graphical representation of the timer-ISR code used for software-to-hardware mapping of the event sequence as executed on the MCU. By default, the routine is triggered at 1 μ s intervals, but the interrupt interval is dynamically adjusted to longer values during data acquisition events. An overall counter, counter_micros, keeps track of the total number of microseconds elapsed. The flow chart shows how the ISR makes numerous tests upon the event data array to distinguish between GPIO switching and ADC acquisition instructions, and to detect when events (or the entire sequence) should terminate. To facilitate the reader's interpretation, it is recommended to first read through the actions following negative test results, i.e., where logical tests return "no".

of the main processor loop to execute the ISR's code instructions. At the same time the counting process starts over again — from zero. The cycle continues indefinitely until a code decision within the ISR or a user intervention halts the timer.

The timer-ISR combination is the standard way to run GPIO and other hardware control processes in embedded systems with CPU-clock-cycle timing precision. On the Teensy 4.1, the NMRduino firmware sets a timer clock source of 150 MHz so that the event sequence time base is generated every 150 count cycles — in other words so that the ISR is triggered every 1 μ s. The Teensy 4.1 in fact has several timers, and the one used here is one of the so-called Quad Timer Modules (QTM), which can be used to produce a time-synchronized output across four channels. One of the channels is configured to generate the ADC conversion trigger pulse for the LTC1859. By default the conversion is triggered every 10 μ s starting from the first ISR (to give a 100 kps sampling rate), although it is permitted to change the trigger period during the ISR – on-the-fly – to allow for dynamic updating of the ADC sampling rate. This background use of timer output ensures

that the data conversions begin at precisely the desired moment, with minimum jitter.

The specific task of the ISR code is to adjust the microcontroller GPIO and peripherals according to the instructions of the event sequence array lists. As such, it is designed in a highly structured way to parse and execute the acquisition/pulse data correctly, while also minimizing the execution time to avoid being terminated "prematurely" by the subsequent timer interrupt.

Sequence events are handled as follows. The ISR counts the number of interrupts that have occurred, and therefore the number of microseconds ($N_{\mu s}$) that have elapsed since the start of the event sequence. Each interrupt performs a logical comparison between $N_{\mu s}$ and the first element of the event time stamp array. The result of the logical comparison can be false or true:

- A false condition occurs when $N_{\mu s}$ is smaller than the first event timestamp, meaning that we have not reached the first event yet, and the ISR takes no further action except to increment $N_{\mu s}$ by 1;

- A true condition occurs when $N_{\mu s}$ equals the timestamp, and the GPIO pin registers are modified so as to change the output channel status according to the respective clear or set instruction. All timestamp and GPIO set/clear array pointers are then incremented by 1 so that subsequent triggers of the ISR that reach a true result will compare with the second, and then third, and fourth event timestamp elements and so on, and update the GPIO accordingly with those events. A flow chart representation of these actions appears in the first column of Fig. 18.

We note, however, that some parts of the ISR code do unavoidably take longer to execute than the $1\mu s$ interrupt period. For instance using the LTC1859 ADC, data are only available $4.5\mu s$ after a conversion trigger, and fetching the data (16 bits) takes a further $\sim 2\mu s$ using 10 MHz SPI. To accommodate these operations the counter match value is actually increased on-the-fly during the ISR to delay the next interrupt (the typical delay is set to $10\mu s$), and restored to the default $1\mu s$ value on the next interrupt after correcting the ISR counter for the time shift. For a graphical representation of the overall ISR code see the flow chart in Fig. 18.

Certainly, more-efficient ISR codes can be written. The situation where during the ISR the main processor is held busy fetching ADC data is known in technical terms as bus-interconnect bottlenecking, and during this time other tasks must be put on hold. A solution to liberate the main processor for more time in between interrupts is to use direct memory access (DMA) hardware control. The IMXRT chip has a highly capable DMA controller that can perform the SPI transactions and write the data directly to/from the RAM in the background, rather than the ISR. In the future we expect NMRduino to take advantage of DMA to obviate some of the event sequence timing constraints.

Finally, the ISR offers another convenient feature for NMR experiments where all pulse events are executed before data acquisition. After the output events are completed, the ISR automatically acquires the remaining data points specified in the user interface (UI) but not explicitly defined in the event sequence (.seq) file. This eliminates the need for manual entry of acquisition parameters into the .seq file, simplifying the programming aspect. Once all events have been output and data points have been acquired, the ISR deactivates the timer using the logical function in the third row of Fig. 18 and seamlessly proceeds to the next stage of the experiment.

A.4. Data export

During the ISR, the fetched ADC data values are stored in an array located in the MCU's RAM — named `data_array` in Fig. 18. This array serves as a buffer to enable in principle the MCU to perform subsequent processing operations, e.g., oversample averaging, anti-alias filtering. The most straightforward approach, however, is simply to transfer all data back to the UI for off-line processing, via the USB port.

Unlike the pulse upload process, which involves the additional handshaking procedure as a hardware safety mechanism, the data transfer via USB serial does not require such steps. It is sufficient for the UI to perform the most basic verification check: ensuring that the number of ADC data points received matches the originally requested number. In the event of a mismatch, the UI promptly aborts the measurement process. In our experience these occurrences are extremely rare thanks to the inherent handshaking and error-catching routines of the USB, and NMRduino consistently achieves error-free data transfers.

Appendix B. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jmr.2024.107665>.

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